

CS 189/289

Some important & timely applications of AI in biology:

1. Protein structure prediction
2. Protein design

CS 189/289

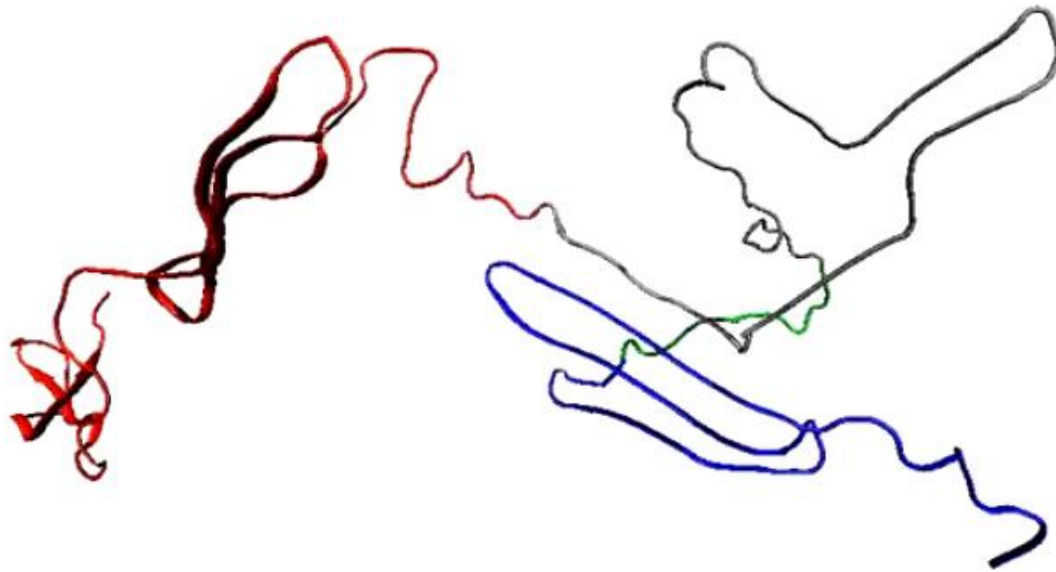
Some important & timely applications of AI in biology:

1. Protein structure prediction
2. Protein design

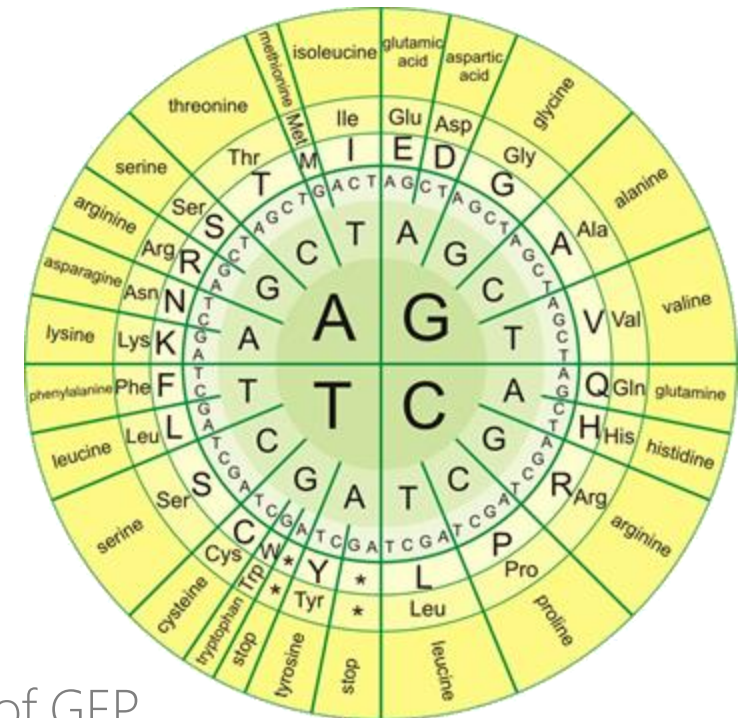
Proteins are strings of nucleotides/amino acids

238 length amino acid sequence:

MSKGEELFTGVVPILVELDGDVNGHKFSVSG
EDFFKS...NSHN**VYIMADKQKNGIKVNF****KIRH**



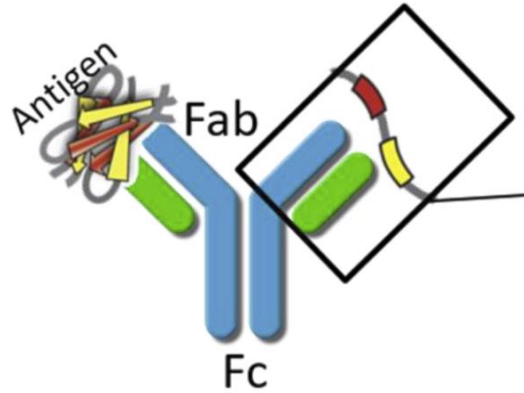
Green fluorescent protein
(GFP) folding itself



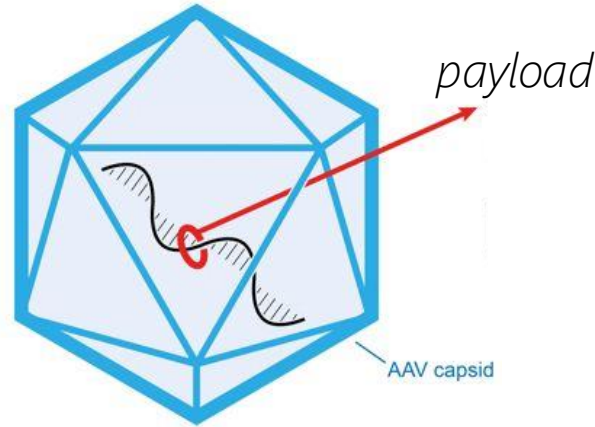
[2008 Nobel in chemistry for discovery and development of GFP,
Osamu Shimomura, Martin Chalfie and Roger Y. Tsien]



Protein engineering: therapeutics, environment, *etc.*



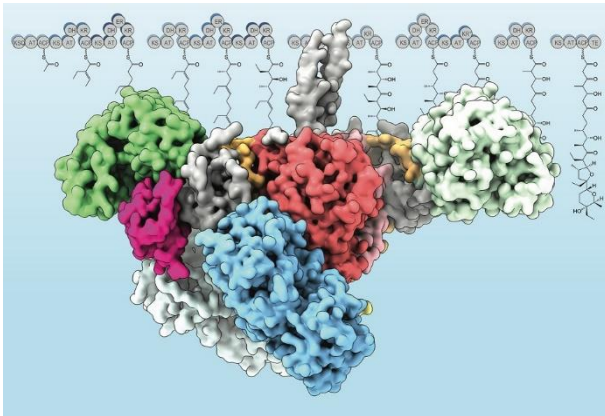
antibody therapeutics



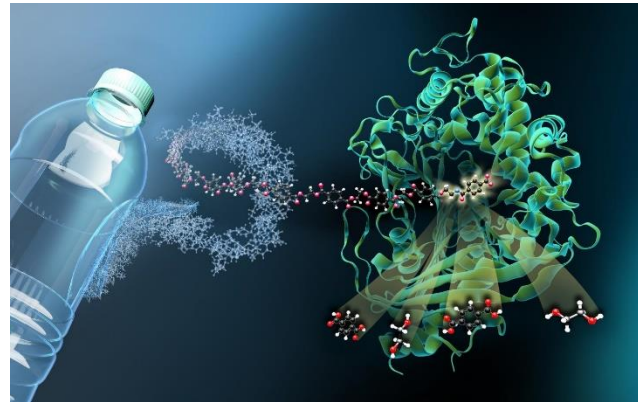
gene therapy virus delivery (AAV)



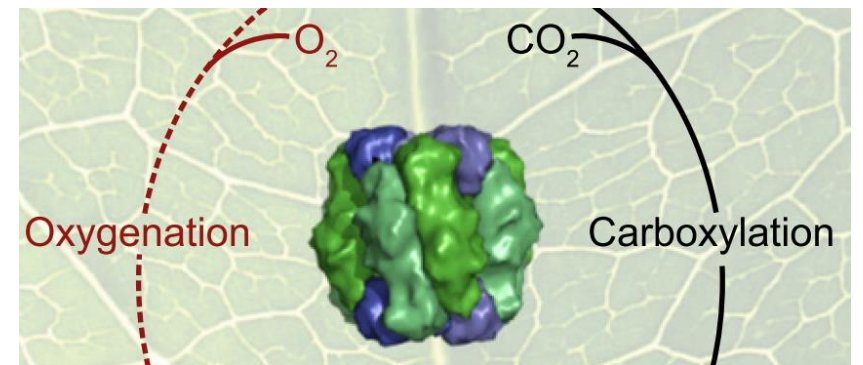
gene editing (CRISPR/Cas9)



antibiotics & biofuel production (PKS)



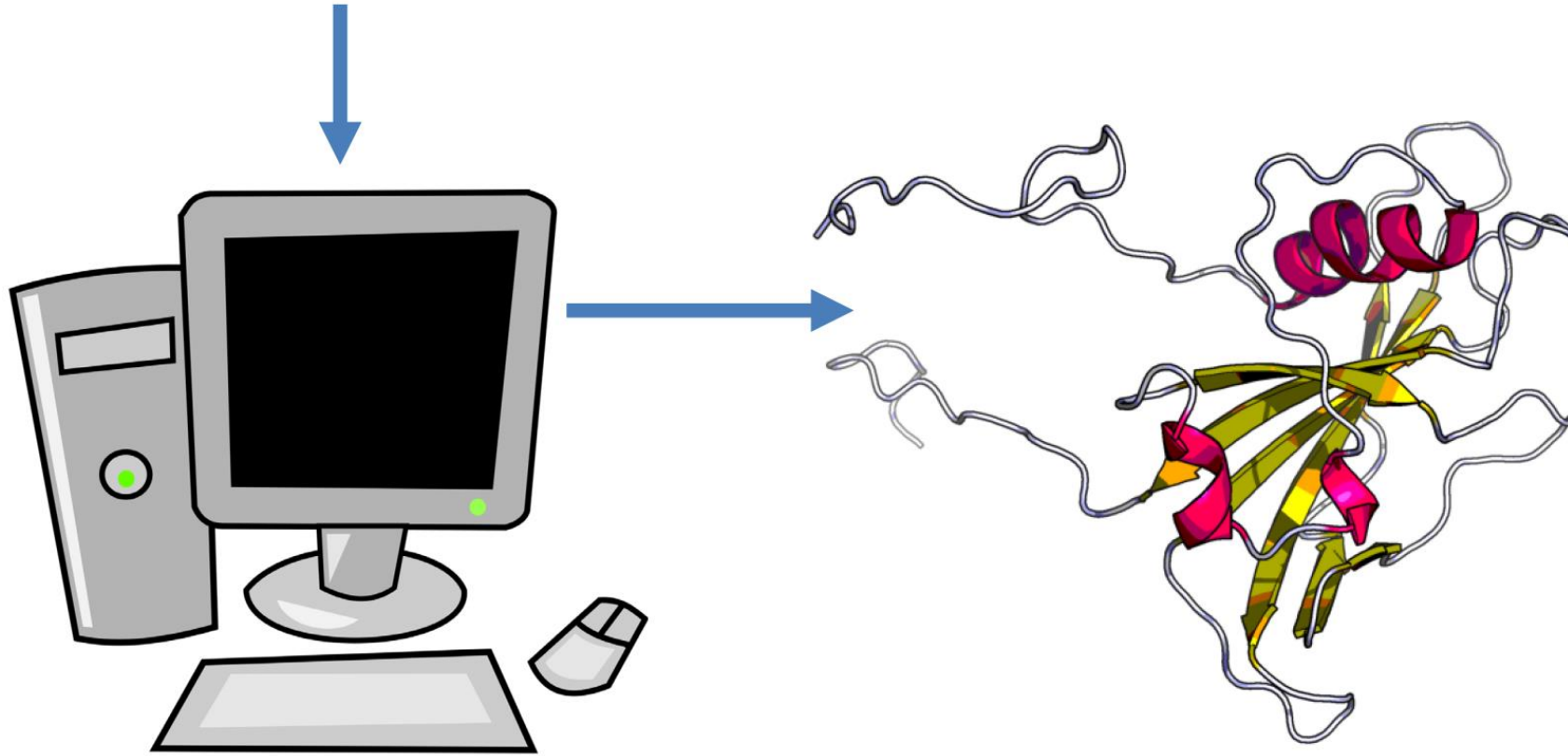
plastic recycling (PETase)



CO₂ biosequestration (RuBisCO)

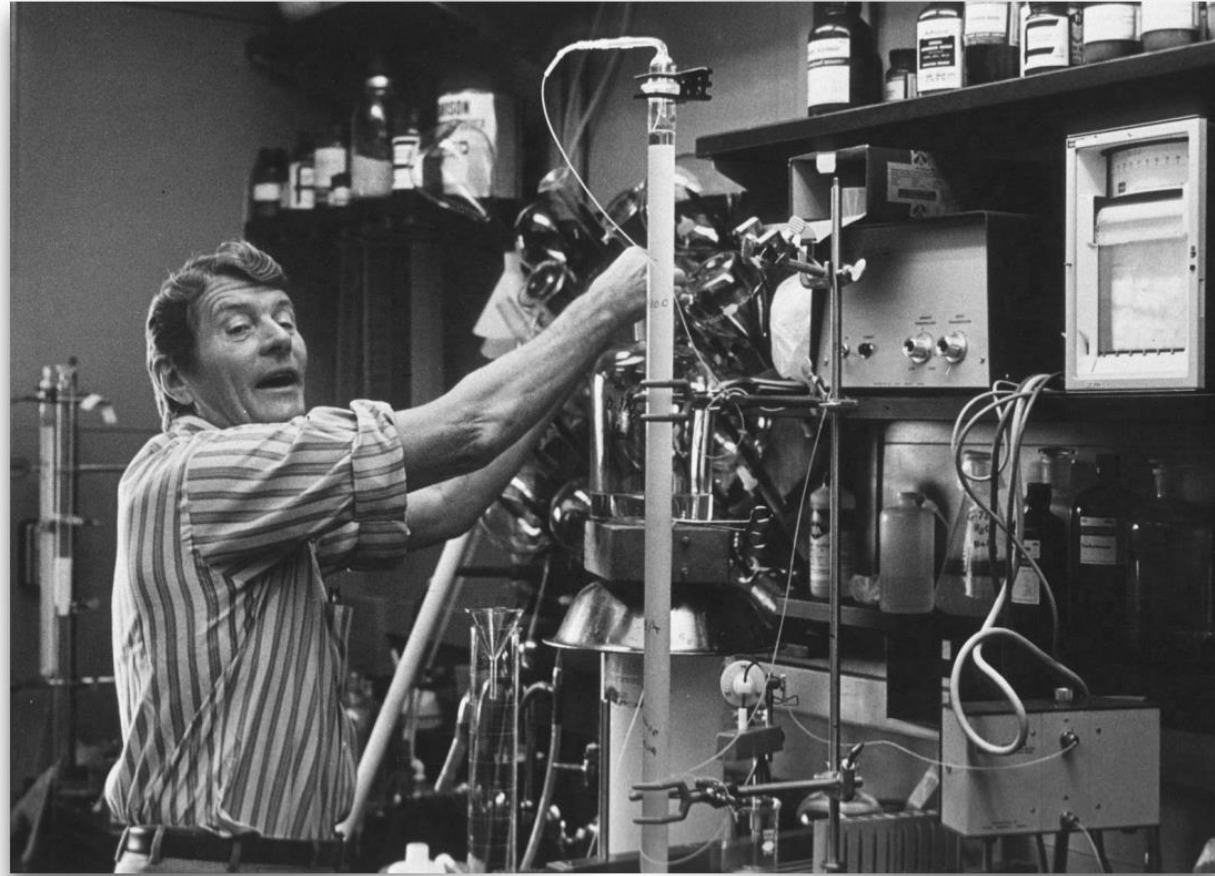
Protein Structure Prediction

MEKVNFLKNGVLRLPPGFRFRPTDEELVVQYLKRKVFSFPLPASIIPEVEVYKSDPWDLPGDMEQEKYFFSTK
EVKYPNGNRSNRATNSGYWKATGIDKQIILRGRQQQQQLIGLKKTLVFYRGKSPHGCRTNWIMHEYRLAN
LESNYHPIQGNWVICRIFLKKRGNTKNKEENMTTHDEVNRNREIDKNSPVVSVKMSSRDSEALASANSELKK



Has been studied several decades

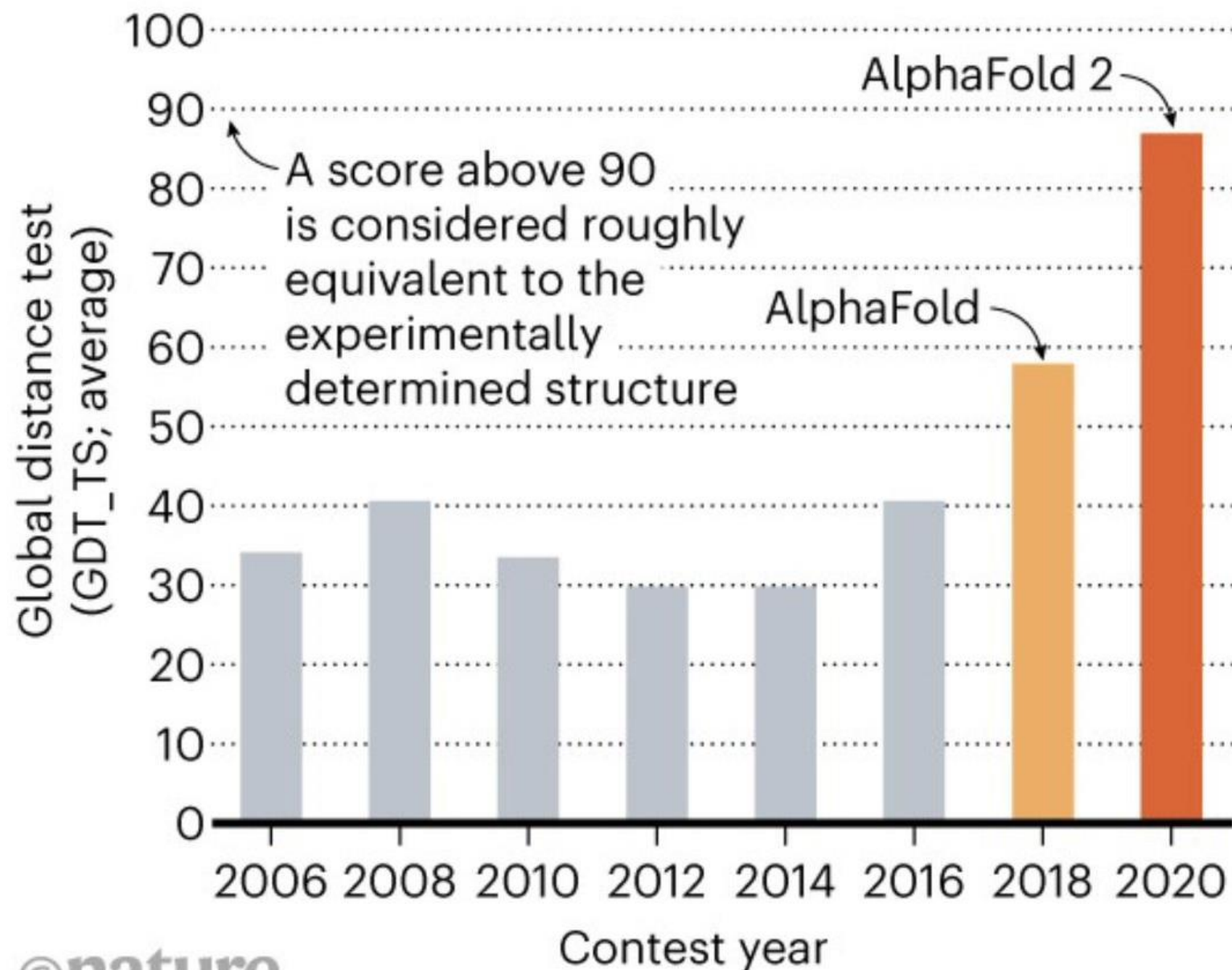
Amino acid sequence determines protein 3D structure



Christian Anfinsen
Nobel Prize in Chemistry 1972

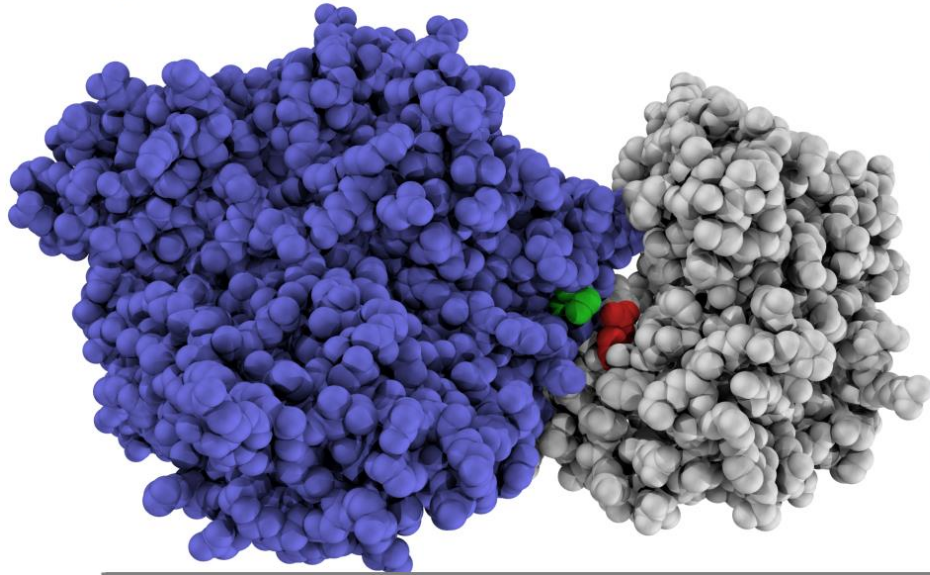
2020

For the first time, state-of-the-art is deep learning based:

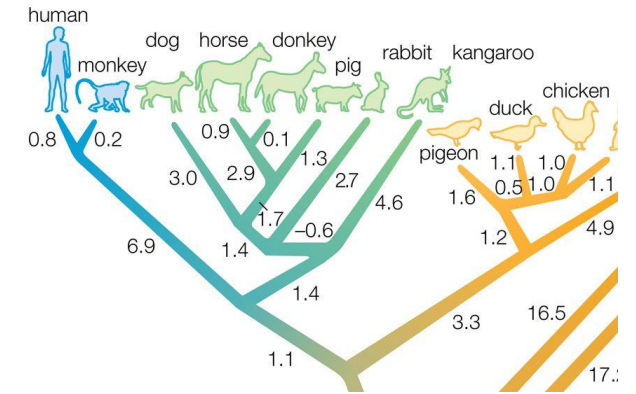


AlphaFold2 relies on previous key insight

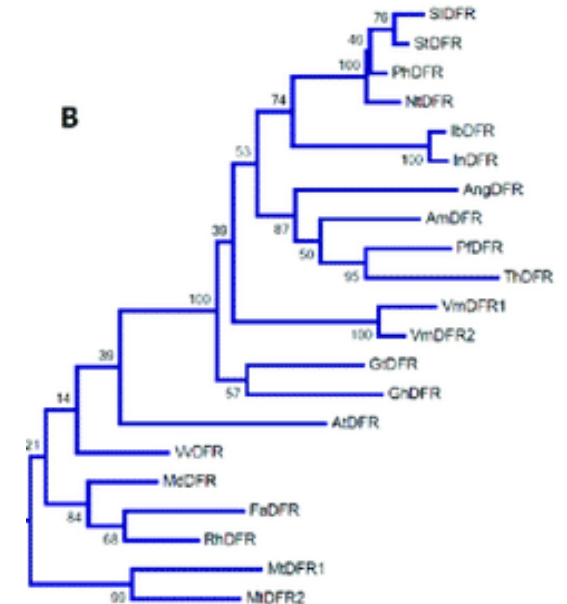
Amino acids in direct physical contact tend to covary or “coevolve” across related proteins



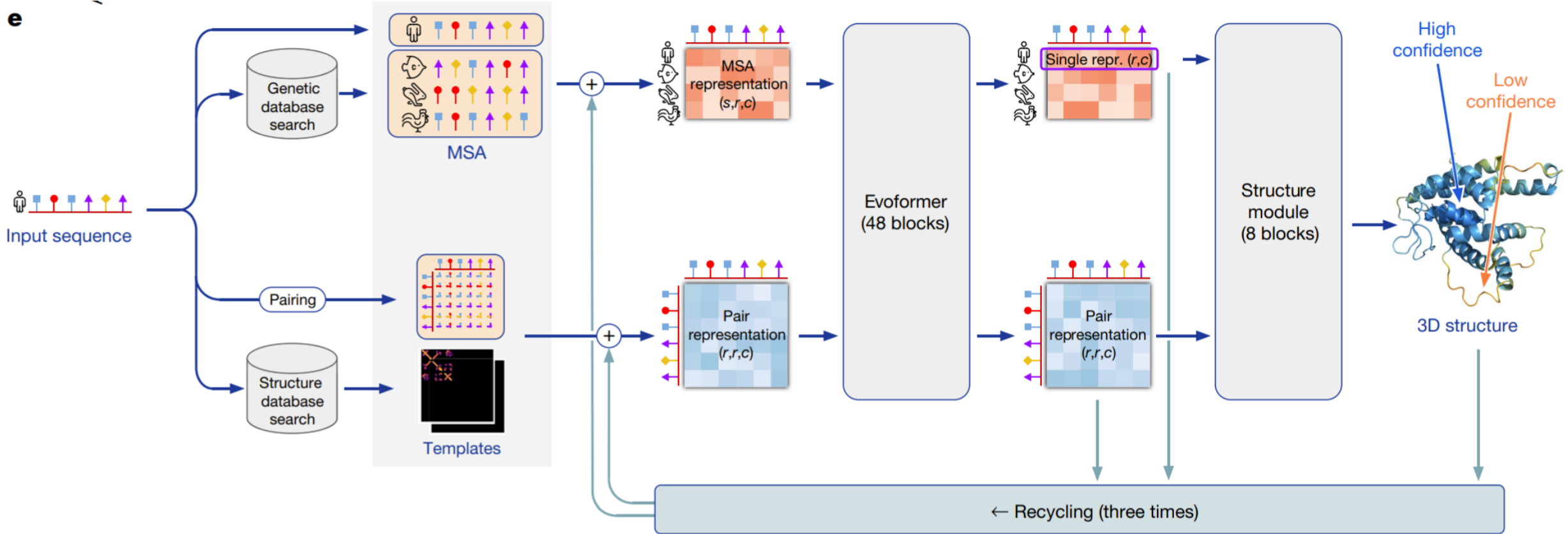
For example, a mutation that causes one amino acid to get bigger is more likely to preserve protein structure and function (and thus survive) if another amino acid gets smaller to make space



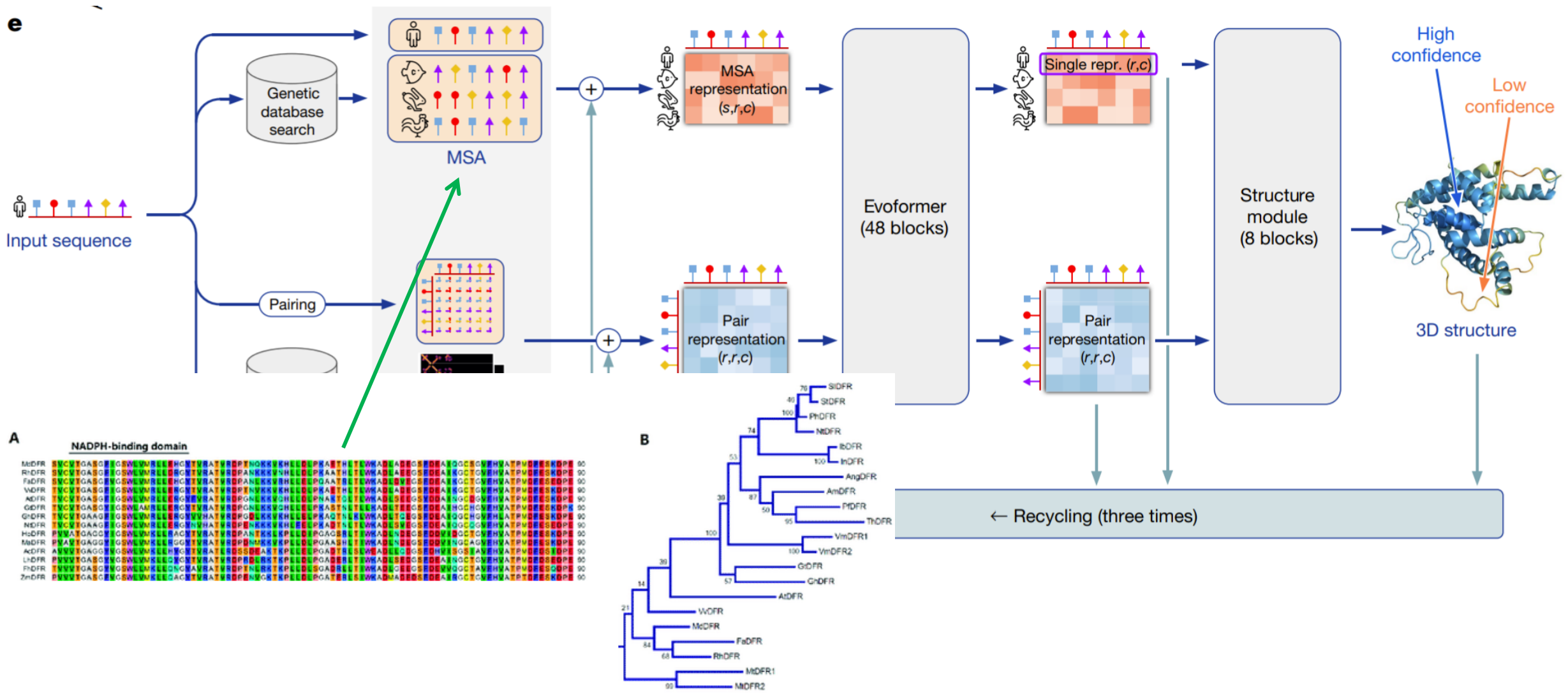
B



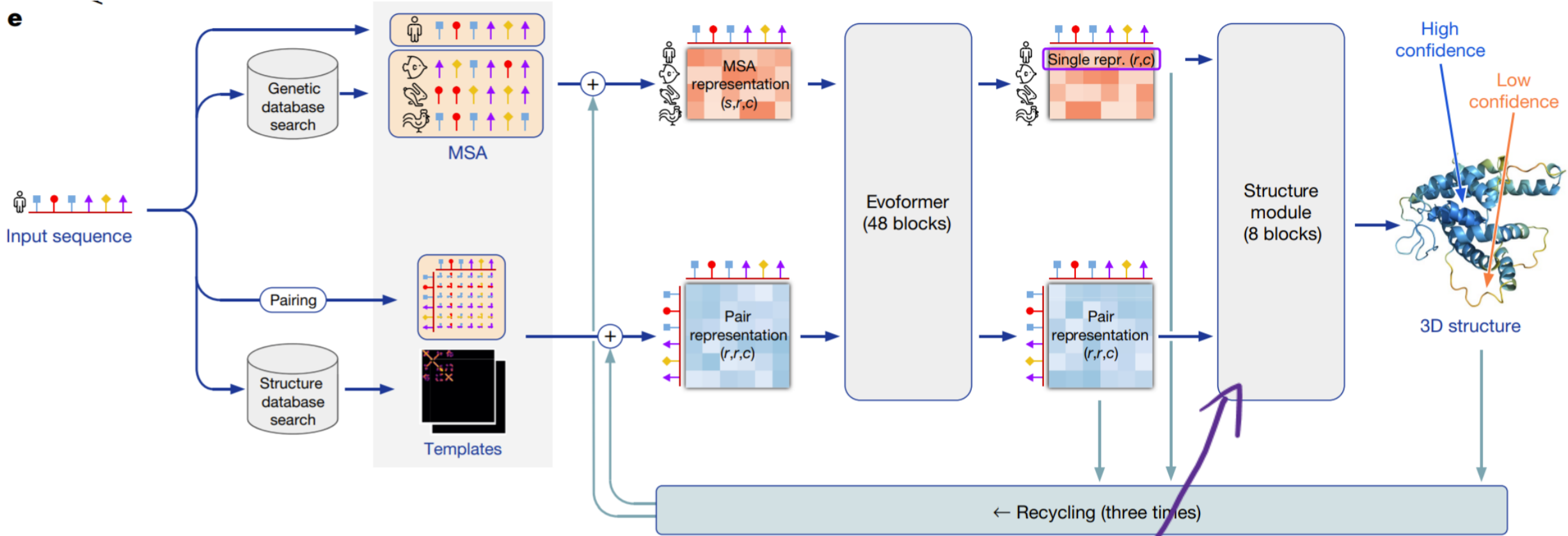
AlphaFold2 “almost end-to-end” neural network



AlphaFold2 “almost end-to-end” neural network



AlphaFold2 “almost end-to-end” neural network



Uses a rotation equivariant attention architecture in structure module

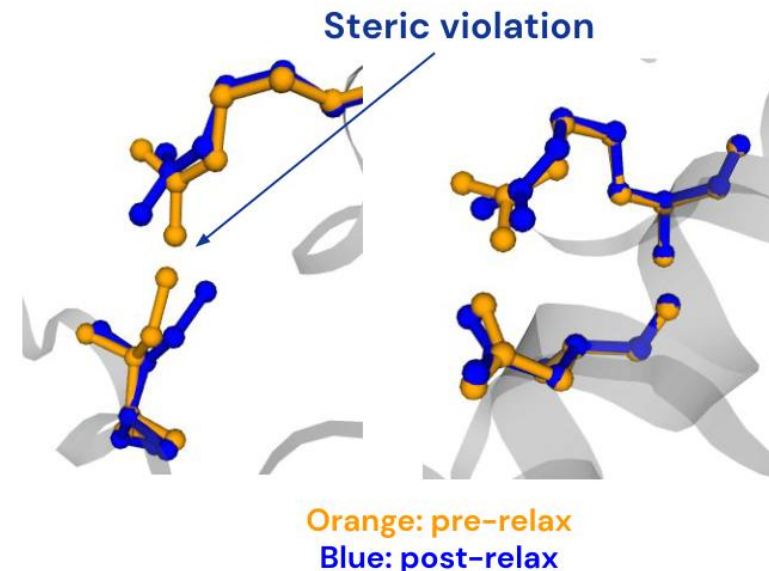
AlphaFold2 “almost end-to-end” neural network

- Can end up with atom positions in violation of physics.
- Thus relies on old style energy-based approaches to refine the predicted 3D coordinates.

Relaxation

© 2020 DeepMind Technologies

- The end result of iterative refinement is not guaranteed to obey all stereochemical constraints
- Violations of these constraints are resolved with coordinate-restrained gradient descent
- We use the Amber ff99SB force field¹ with OpenMM²



Some thoughts on AlphaFold2

- DeepMind took on a long-tackled, well-defined problem, with clear data, clear benchmarks, and a clear way to demonstrate improvement.
- Expense of protein structure data used for AlphaFold2, conservatively estimated at ~US\$20 billion (Burley et al., 2023).
- They relied heavily on years of prior work in protein folding research: “template-based modelling”, “evolutionary co-evolution modelling”, “contact prediction”, energy-functions.

The perpetual motion machine of AI-generated data and the distraction of ChatGPT as a ‘scientist’

[Jennifer Listgarten](#) 

[Nature Biotechnology](#) (2024) | [Cite this article](#) | Correspondence | [Published: 25 January 2024](#)



Video talk: “NAS 161st Annual Meeting – Symposium AI and Scientific Discovery”

CS 189/289

Some important & timely applications of AI in biology:

1. Protein structure prediction
2. Protein design

PROTEIN ENGINEERING

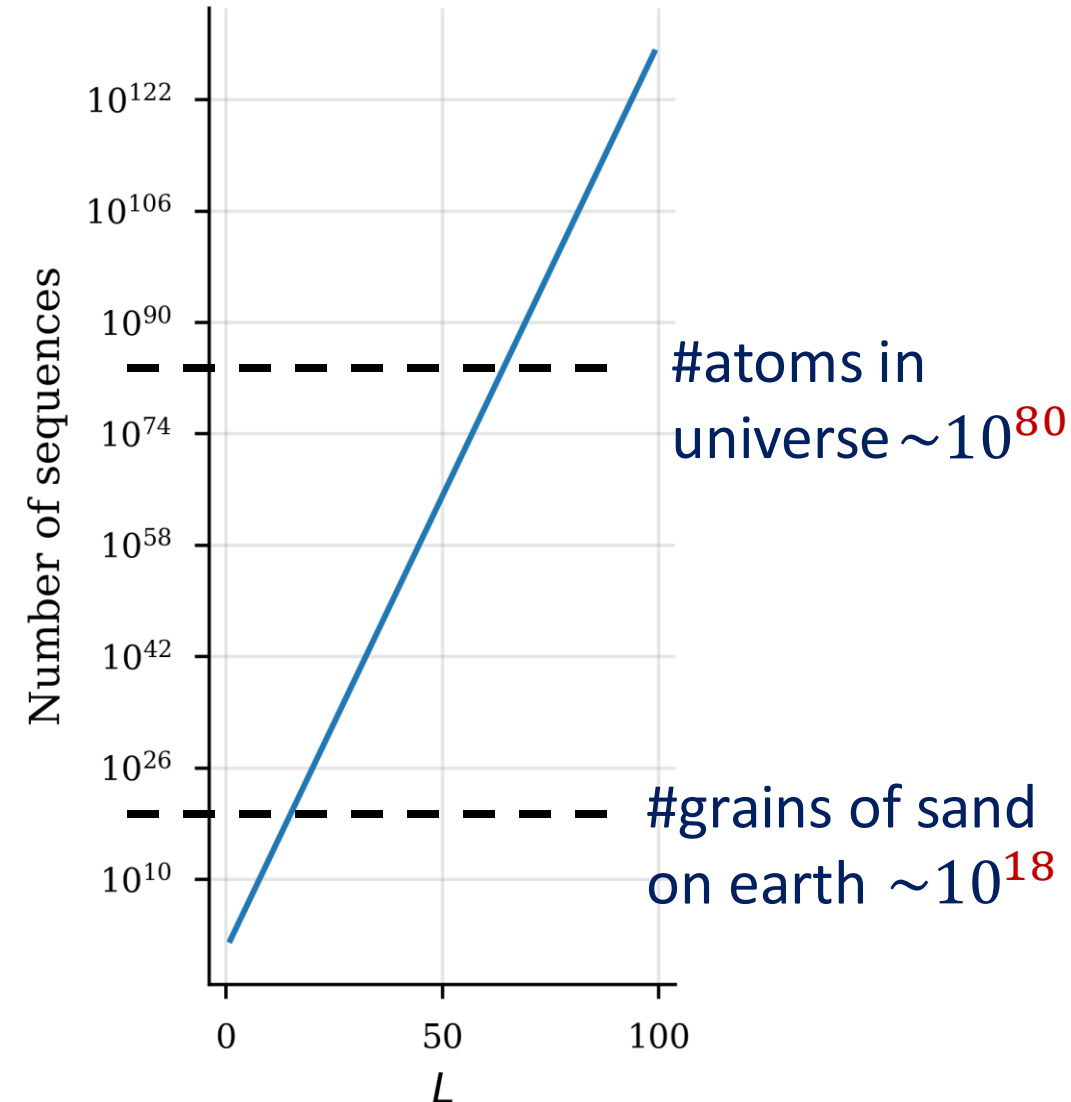
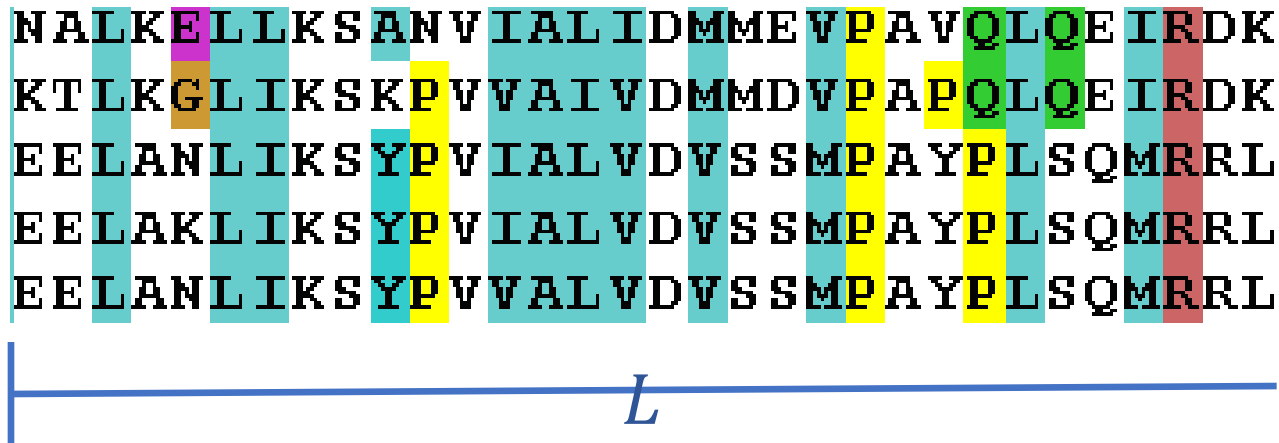
Science
AAAS

**How artificial intelligence is
reengineering protein engineering**

Jennifer Listgarten^{1,2,3*} and Hanlun Jiang¹

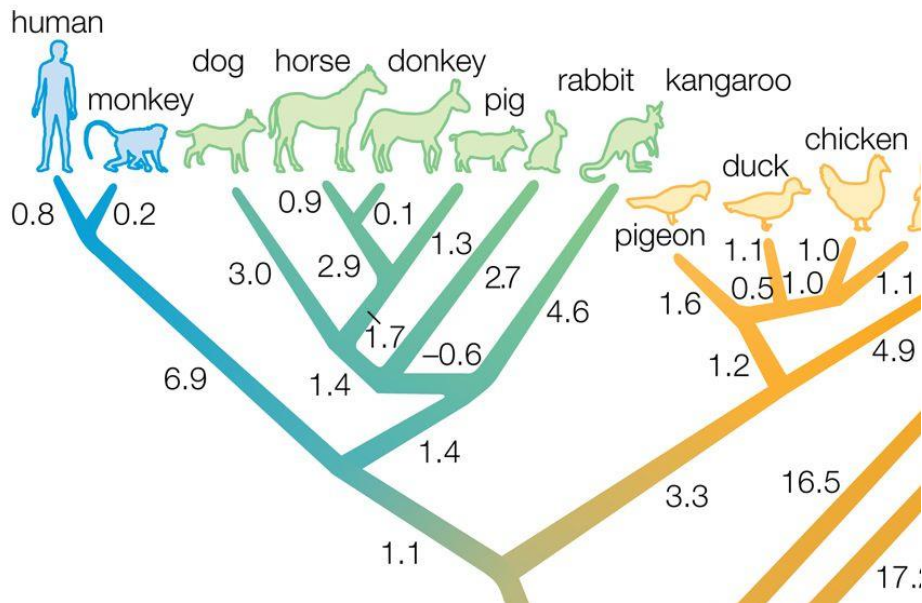
Fundamental difficulty: design space is nearly infinite

- Also highly rugged design space
⇒ size scales as $\sim 20^L$
- Discrete search space (no gradients)

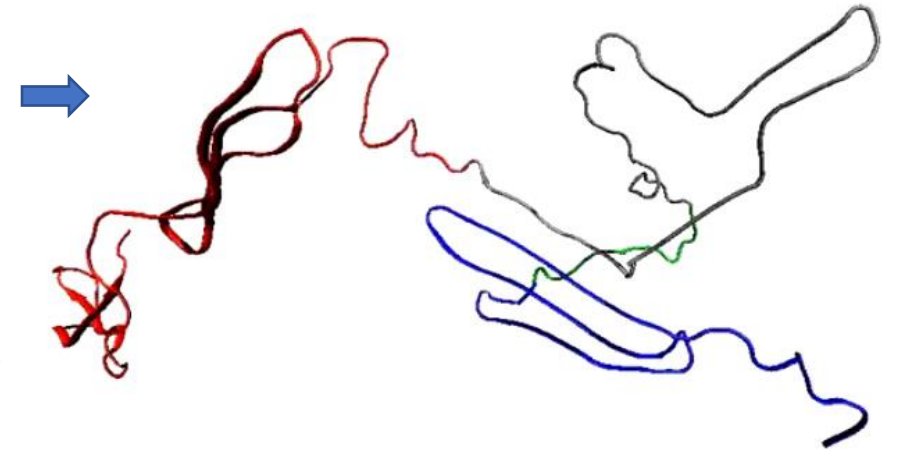


Successes in navigating this complex space

1. Nature: via *evolution* over *millions of years*.



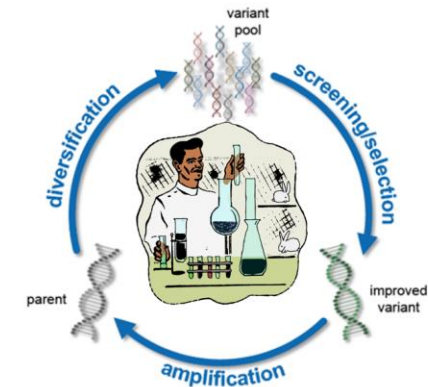
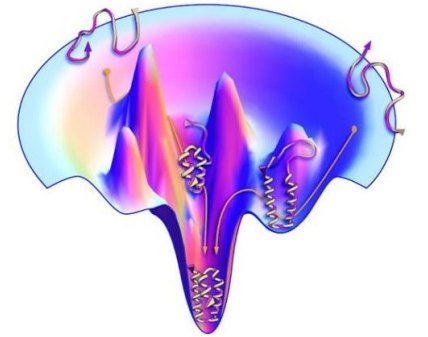
MSKGEELFTGVVPILV
ELDGDVNGHKFSVSG
EGEGDATYGKLTCLKFI
CTTGKLPVPWPTLVTT
FSYGVQCFSRYPDHM
KQHDFFKSAMPEGYV
QERTIFFKDDGNYKTR
AEVKFEGDTLVRIELK
GIDFKEDGNILGHKLEY
NYNSHNVYIMADKQKN
GIKVNFKIRHNIEDGSV
QLADYQQNTPIGDGPV
LLPDNHYLSTQSALSK
DPNEKRDHMLLEFVT
AAGITHGMDELYK



green fluorescent
protein folding itself

Protein engineering strategies **emerging**

- i. Computation ("data free"): **physics-based energy functions** (e.g., Rosetta) to model **protein structure**, and protein binding.
~1997-2023'ish (almost R.I.P.) [2024 Nobel Prize]
- ii. Wetlab: **directed evolution** to iteratively directly design property of interest.
~1993-present [2018 Nobel Prize]
- iii. Machine learning (augmented): generative models; function prediction; structure prediction, etc. ~2018(?) - present



Did AlphaFold2/3 “solve” protein engineering?

NEWS | 22 July 2021

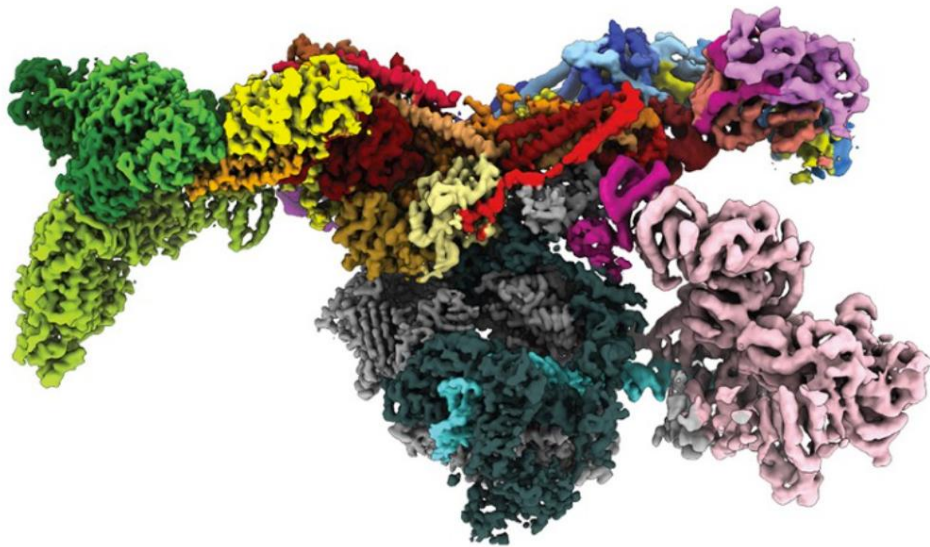
DeepMind’s AI predicts structures for a vast trove of proteins

AlphaFold neural network produced a ‘totally transformative’ database of more than 350,000 structures from *Homo sapiens* and 20 model organisms.

[Ewen Callaway](#)



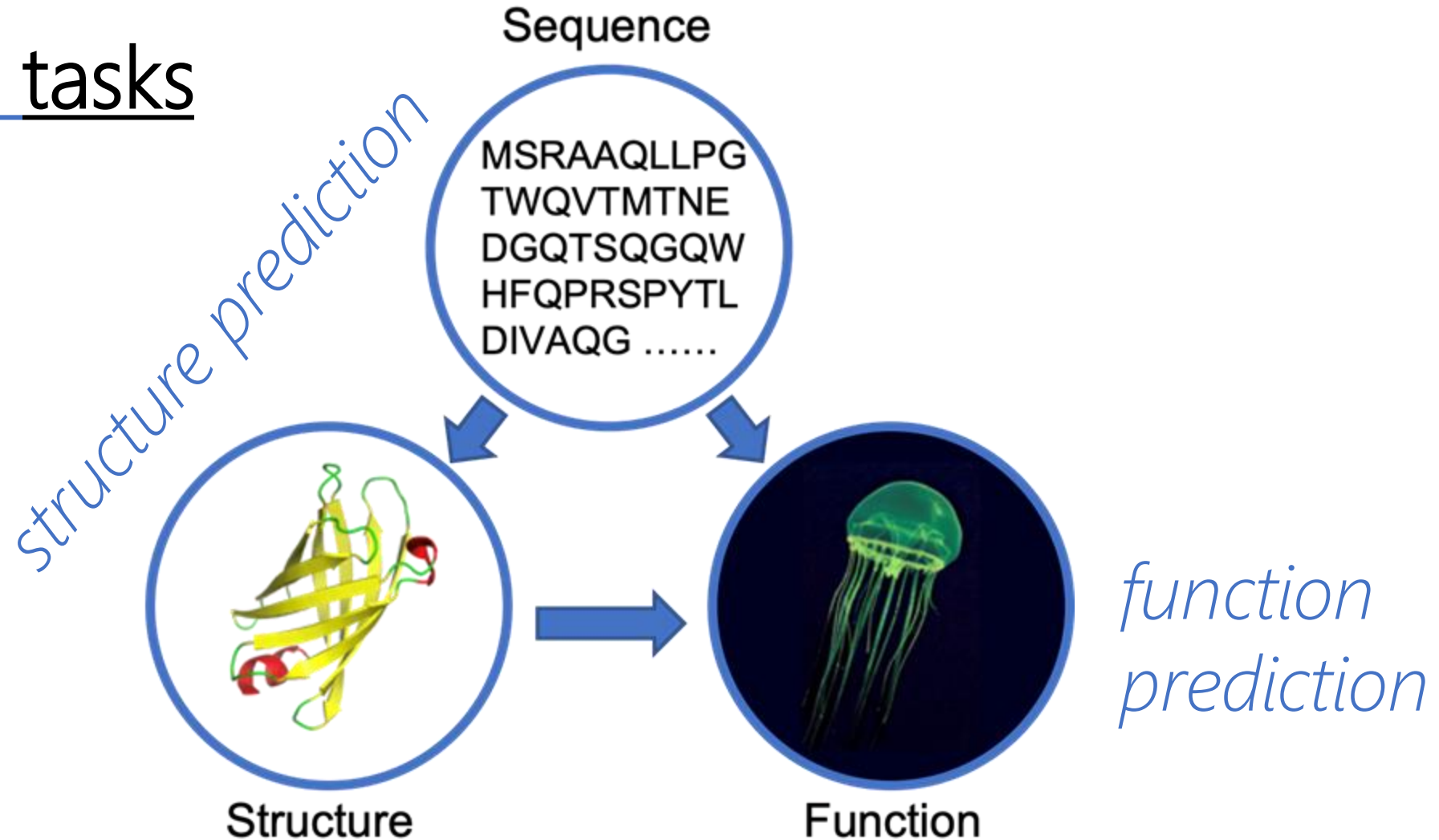
sequence → *structure*



- No: don’t typically know which protein structures we need.
- If did, would need: *structure* → *sequence*.
(decent ML solutions exist).
- Bottleneck challenge: **predict** which proteins have the **function** we desire—**often extrapolatively**.
- AlphaFold2 *was* a breakthrough, and is already useful.

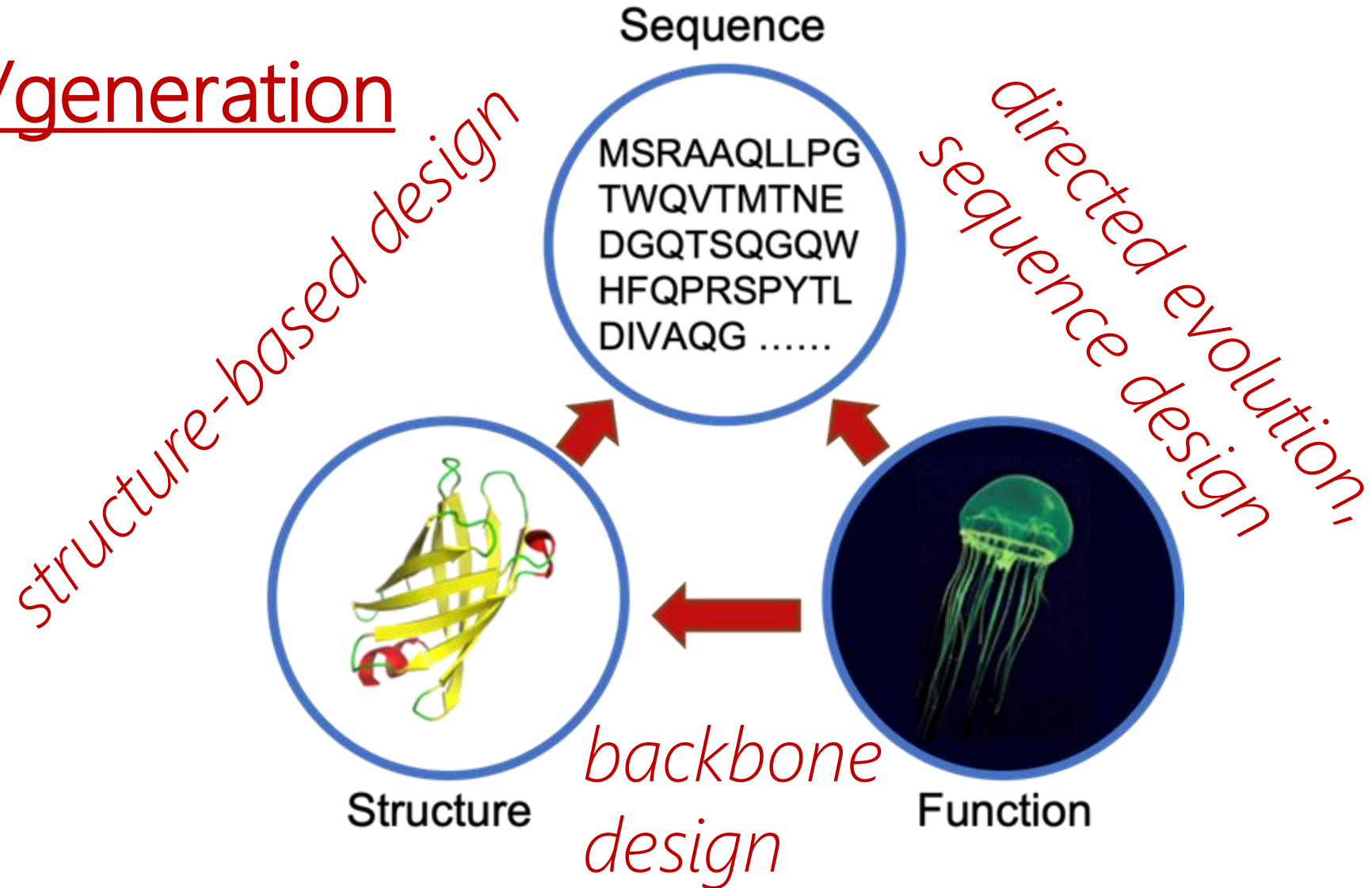
A suite of ML protein engineering problems

Prediction tasks



A suite of ML protein engineering problems

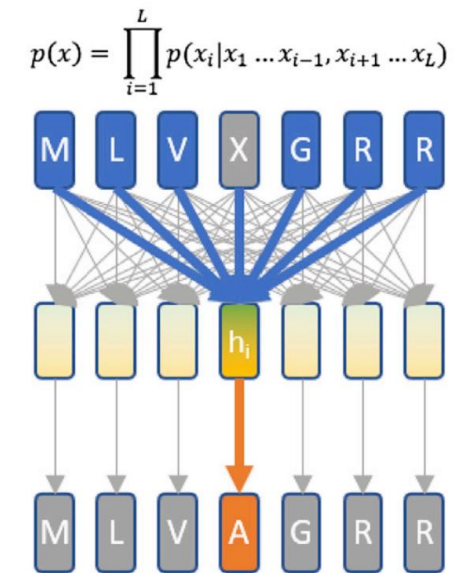
Design/generation tasks



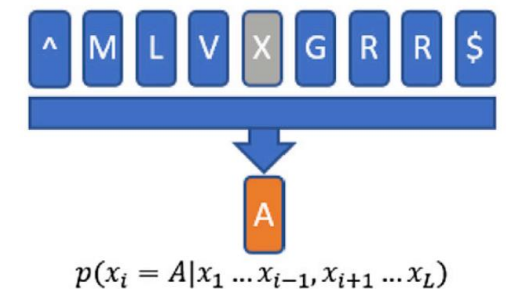
Some trends in ML + protein engineering

1. Representation learning:
un(self)supervised learning on large-scale databases (millions of natural proteins, with e.g., Transformers), or families with Potts/VAE.

[This is really (approx.) density estimation, $p_{\theta}(\text{sequence})$.]



Processes whole sequence

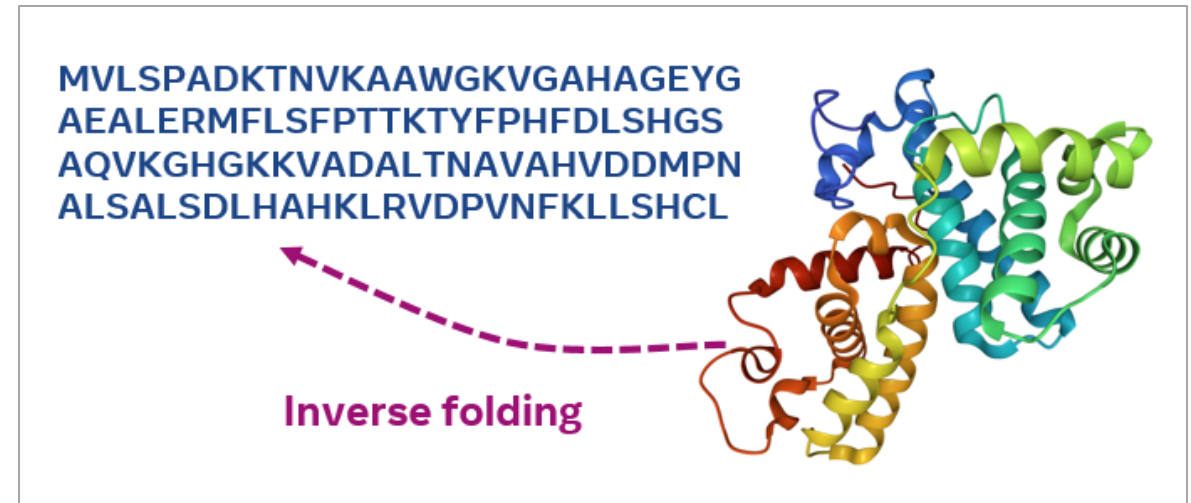


[Bepler et al., Cell Systems 2021]

Some trends in ML + protein engineering

2. (Conditional) generative models for sequences.

- a) structure-conditioned,
aka “inverse folding”
- b) “control tag” conditioned,
e.g., protein family



[This is really (conditional) density estimation and sampling,
 $\text{seq} \sim p_{\theta}(\text{seq} | \mathbf{C})$, *e.g.*, auto-regressive Transformer, diffusion/flow.]

Some trends in ML + protein engineering

3. (Conditional) generative models for structure.

- On (e).

- Pair **Generative models for protein structures and sequences**

[Chloe Hsu](#) , [Clara Fannjiang](#) & [Jennifer Listgarten](#) 

[Nature Biotechnology](#) **42**, 196–199 (2024) | [Cite this article](#)

x_1

[Ingraham et al. Nature 20223]

x_t

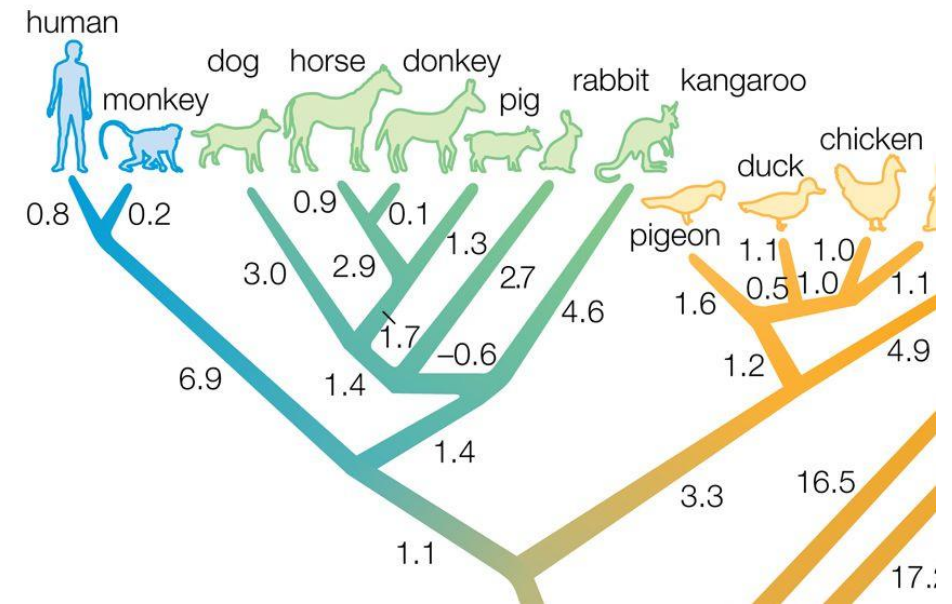
x_0

- [This is really (conditional) density estimation and sampling, $\text{backbone} \sim p_\theta(\text{backbone}|\mathbf{F})$, (e.g. diffusion/flow matching).]

Some trends in ML + protein engineering

4. ML to estimate function from sequence and/or function:

- Few or no labelled data ("zero"/"few" shot learning).
- *Leverage evolutionary information**, or large unsupervised models on pan-proteomic database.



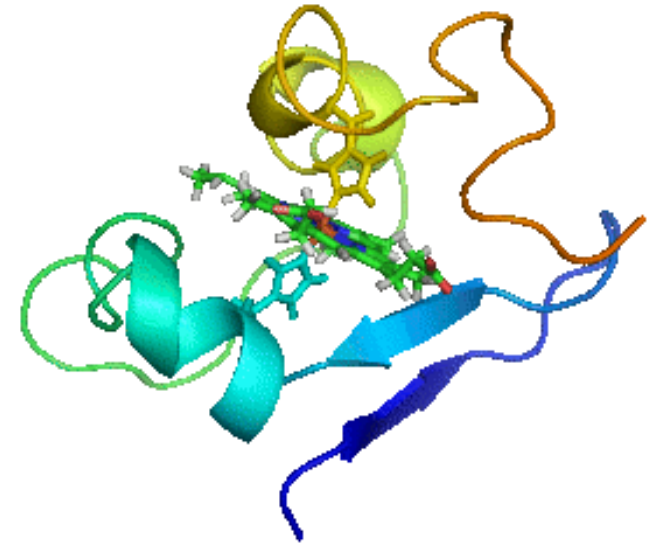
**key part of AlphaFold*

[e.g., $p_{\theta}(\text{function}|\text{sequence})$, $p_{\theta}(\text{sequence}|\text{MSA})$]

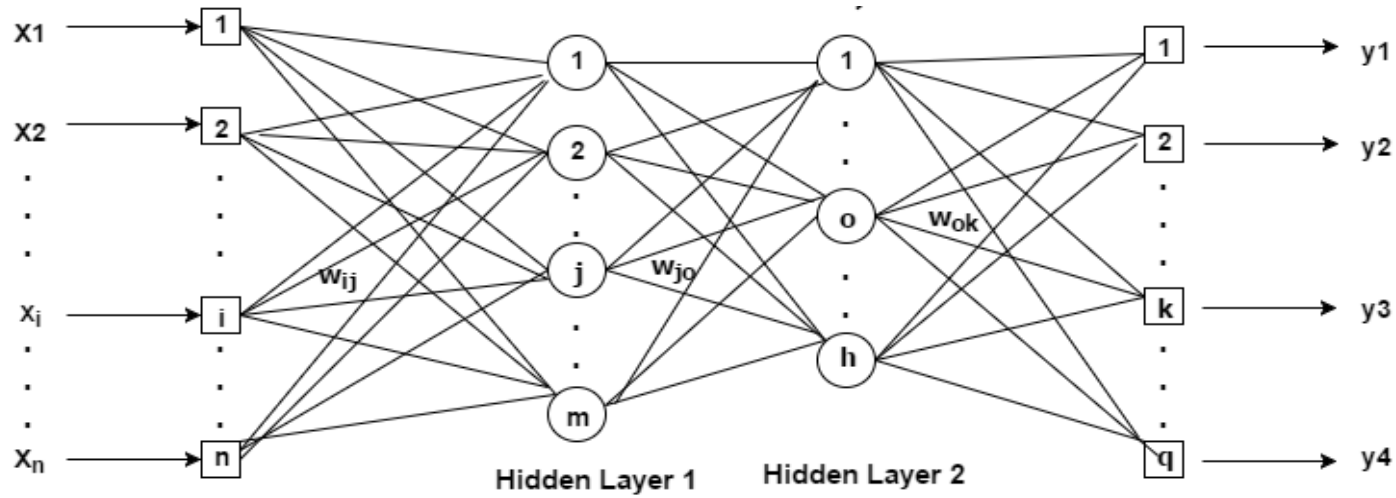
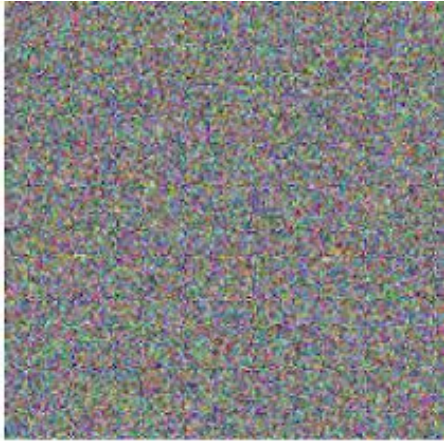
Some trends in ML + protein engineering

5. Structure prediction: filling the gaps left by AlphaFold

- Orphan proteins (with no/few homologs).
- Protein **dynamics and conformational distributions**.
- Protein-protein/DNA/RNA/small molecule binding.



Analogy: can we trust "banana" design?

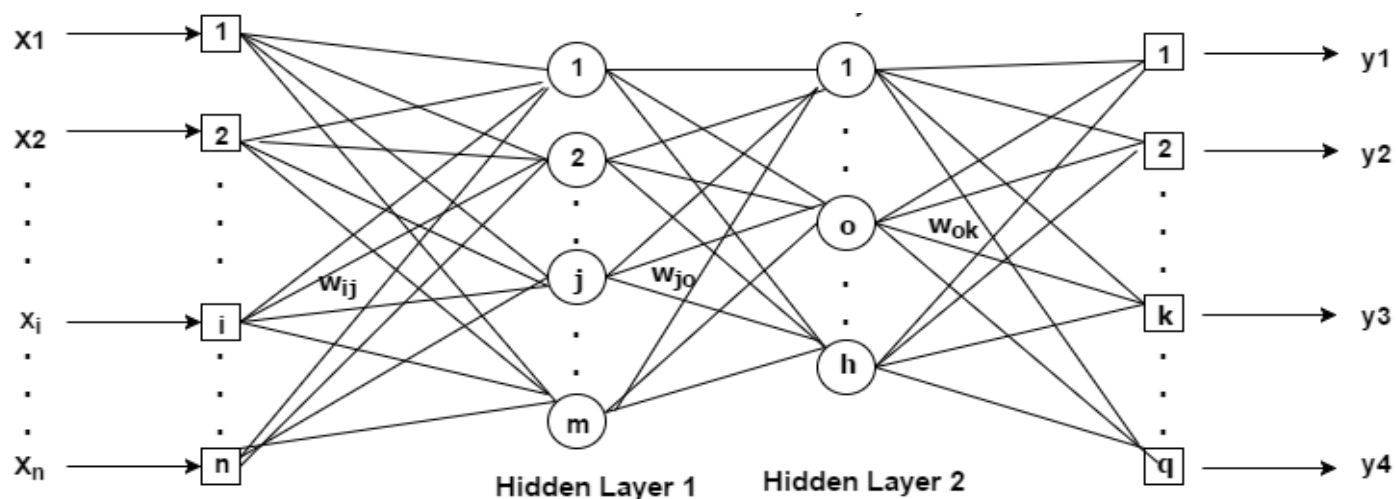


desired property

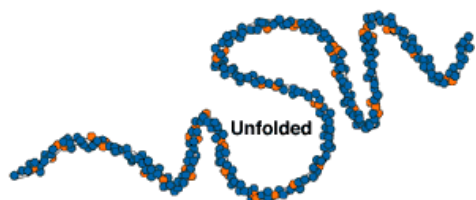


catalytic
efficiency ↑

Naïve design yields abstract art ("pathology-finding").



desired property

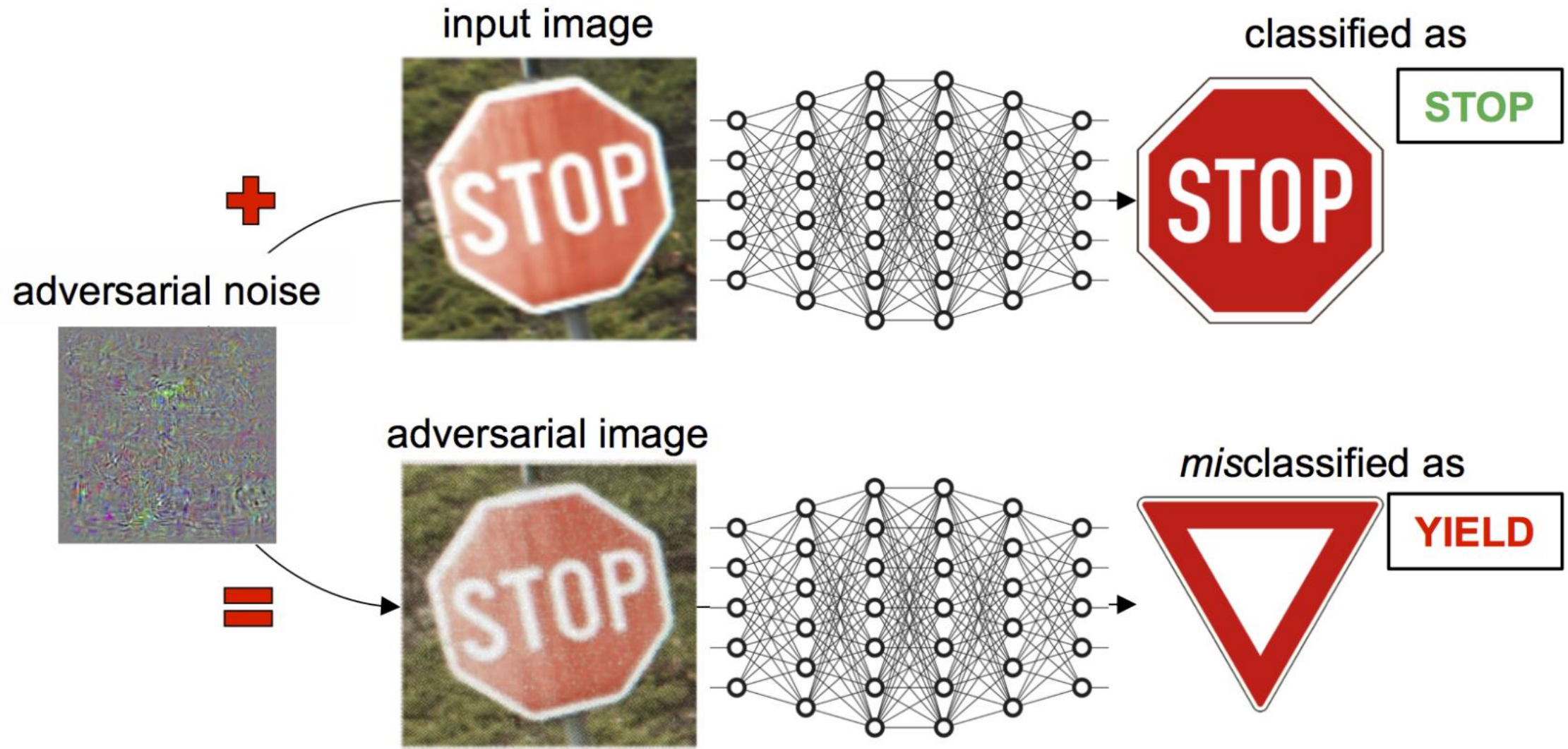


non-folding protein

catalytic
efficiency ↑

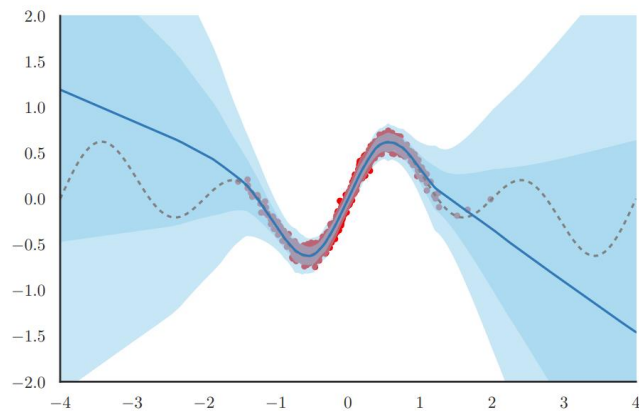
1. Brookes *et al* ICLM 2019 (CbAS)
2. Fannjiang *et al* NeurIPS 2020 (autofocus)

Pathologies of DNNs: in design, we're the adversary

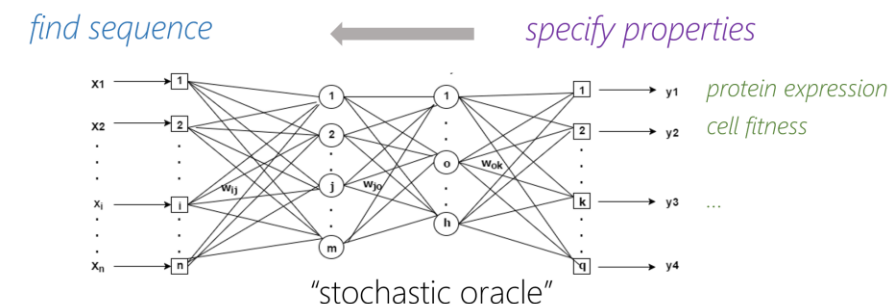


ML-based design challenges tackled in our group

1. A **natural tension** between leveraging the trained model for **extrapolation**, vs knowing that the model is **not trustworthy** in many areas of protein space (related to causality) [1,2].

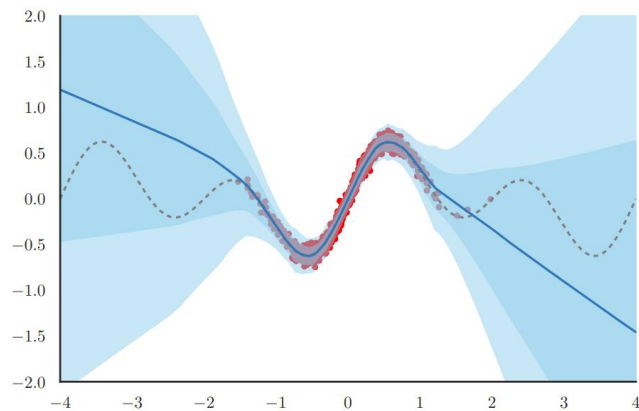


1. Brookes *et al* ICLM 2019 (CbAS)
2. Fannjiang *et al* NeurIPS 2020 (autofocus)

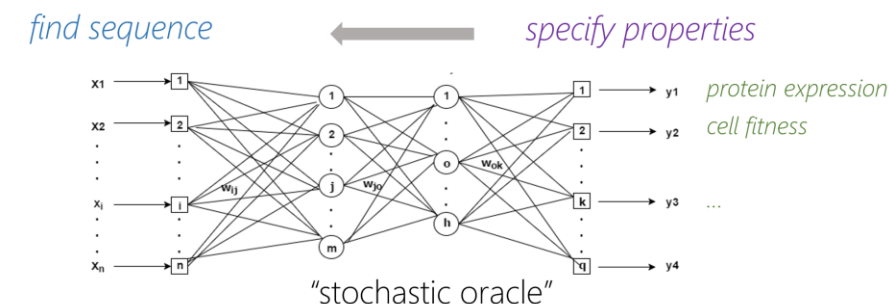


ML-based design challenges tackled in our group

1. A **natural tension** between leveraging the trained model for **extrapolation**, vs knowing that the model is **not trustworthy** in many areas of protein space (related to causality) [1,2].
2. Also related to estimation of **epistemic uncertainty** (whereas we typically think mostly of *aleotoric*) uncertainty [3, 4].

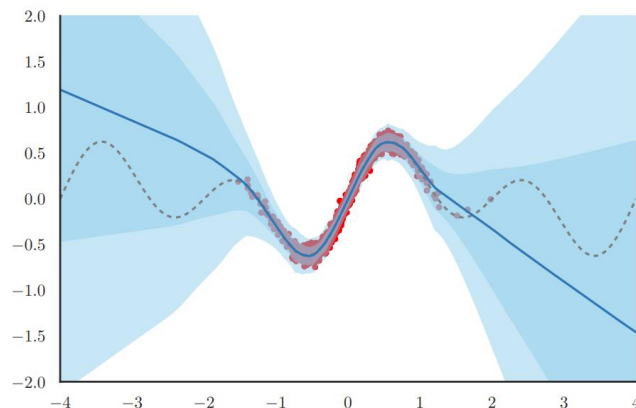


1. Brookes *et al* ICLM 2019 (CbAS)
2. Fannjiang *et al* NeurIPS 2020 (autofocus)
3. Nisinoff *et al* ACS Synth Bio 2023 (fv-BNN)
4. Fannjiang *et al* PNAS 2023(conformal)

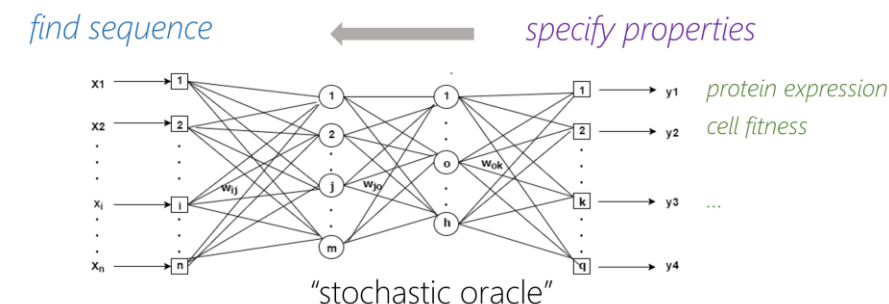


ML-based design challenges tackled in our group

1. A **natural tension** between leveraging the trained model for **extrapolation**, vs knowing that the model is **not trustworthy** in many areas of protein space (related to causality) [1,2].
2. Also related to estimation of **epistemic uncertainty** (whereas we typically think mostly of *aleotoric*) uncertainty [3,4].
3. Suitable protein **inductive biases** when using neural networks [3,5,6,7].



1. Brookes *et al* ICLM 2019 (CbAS)
2. Fannjiang *et al* NeurIPS 2020 (autofocus)
3. Nisinoff *et al* ACS Synth Bio 2023 (fv-BNN)
4. Fannjiang *et al* PNAS 2023(conformal)
5. Aghazadeh *et al* Nat. Comm. 2021
6. Brookes *et al* PNAS 2022
7. Hsu *et al* Nat. Biotech. 2022



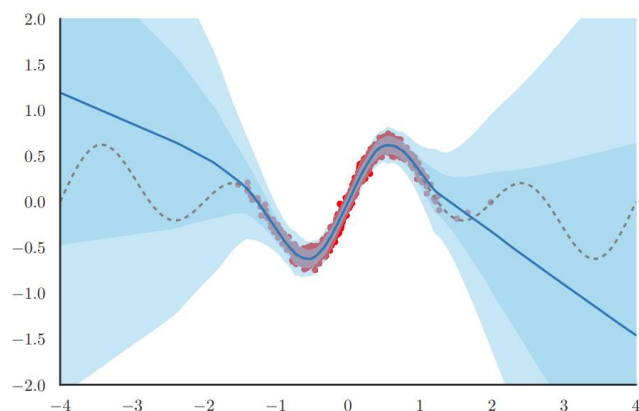
ML-based design challenges tackled in our group

1. A **natural tension** between leveraging the trained model for **extrapolation**, vs knowing that the model is **not trustworthy** in many areas of parameter space [1,2].
2. Also related to the **design of distributions** we typically train on [3,5,6,7].
3. Suitable parameter space for training [3,5,6,7].
4. **Design of distributions** instead of individual sequences [1,2,8].

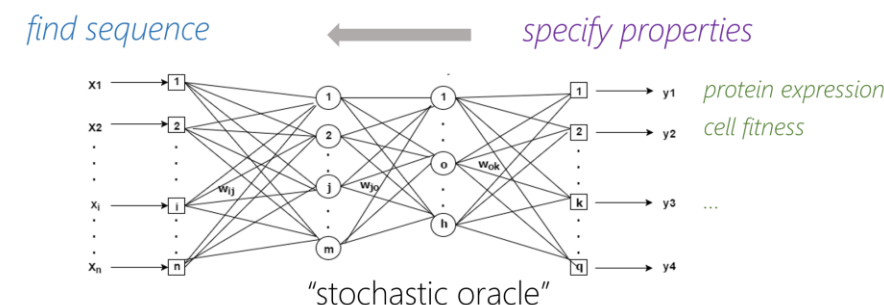
Is Novelty Predictable?

Clara Fannjiang and Jennifer Listgarten

Cold Spring Harb Perspect Biol 2023

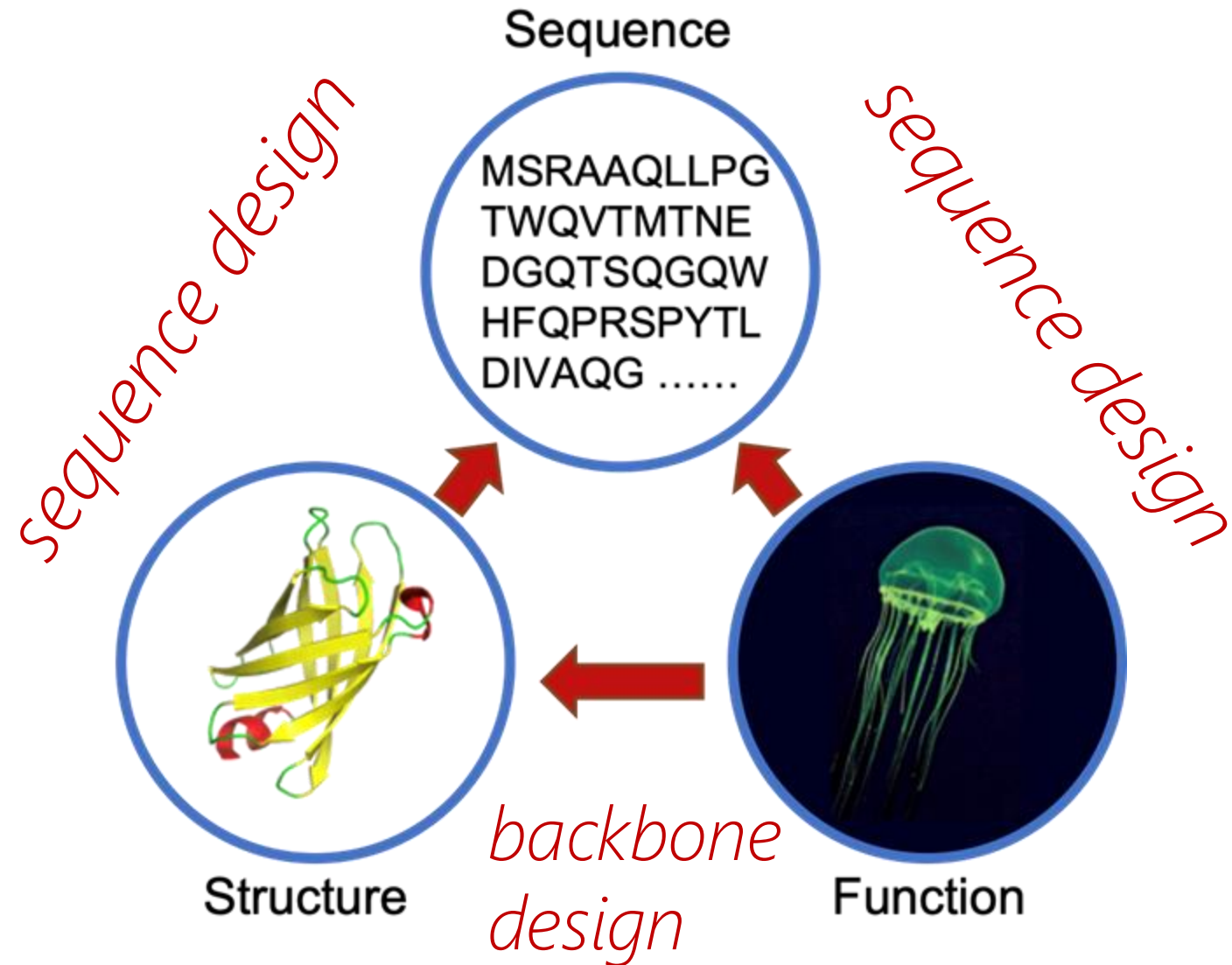


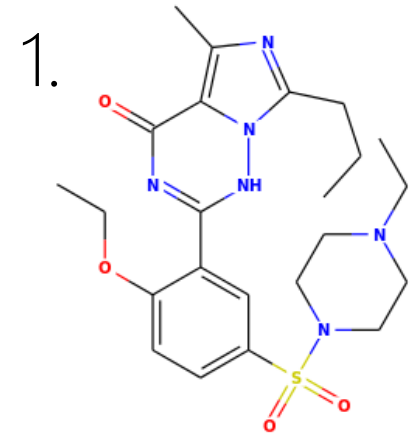
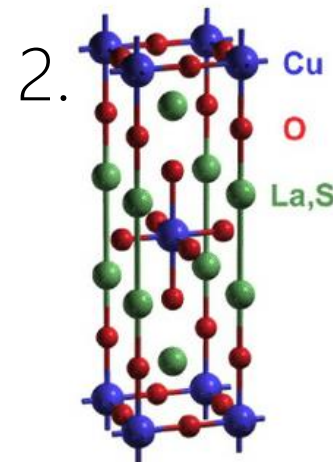
1. Brookes *et al* ICLM 2019 (CbAS)
2. Fannjiang *et al* NeurIPS 2020 (autofocus)
3. Nisinoff *et al* ACS Synth Bio 2023 (fv-BNN)
4. Fannjiang *et al* PNAS 2023(conformal)
5. Aghazadeh *et al* Nat. Comm. 2021
6. Brookes *et al* PNAS 2022
7. Hsu *et al* Nat. Biotech. 2022
8. Zhu, Brookes *et al* Science Advances 2024



Big current focus: sequence generative models

Same technical underpinnings for other discrete objects:



- 
- 

$\text{La}_{2-x}\text{Sr}_x\text{CuO}_4$ (LSCO)
- Natural language generative models like ChatGPT.

How to condition for design: $\boldsymbol{x} \sim \boldsymbol{p}(\boldsymbol{x}|\boldsymbol{y})$?

- Suppose I have a generative model for protein sequences, $\boldsymbol{p}(\boldsymbol{x})$.
 - *e.g.*, from database of protein sequences (*e.g.*, ESM3, ProteinMPNN, *etc.*)

How to condition for design: $\mathbf{x} \sim \mathbf{p}(\mathbf{x}|\mathbf{y})$?

- Suppose I have a generative model for protein sequences, $\mathbf{p}(\mathbf{x})$.
 - *e.g.*, from database of protein sequences (*e.g.*, ESM3, ProteinMPNN, *etc*)
- But I want to generate from a conditional generative model, $\mathbf{p}(\mathbf{x}|\mathbf{y})$, conditioned on property of interest.
 - *e.g.*, condition on $\mathbf{y} = \text{EC\#}$, solubility, expression.

How to condition for design: $x \sim p(x|y)$?

- Suppose I have a generative model for protein sequences, $p(x)$.
 - e.g., from database of protein sequences (e.g., ESM3, ProteinMPNN, etc)
- But I want to generate from a conditional generative model, $p(x|y)$, conditioned on property of interest.
 - e.g., condition on $y = \text{EC\#}$, solubility, expression.
- And I have access to either $\{(x_i, y_i)\}_i$ or a predictive model $p(y|x)$.
 - e.g., your experimental data, predictive model CLEAN¹ for EC#, solubility predictor

How to condition for design: $x \sim p(x|y)$?

Three (statistically correct) ways to do this:

1. Start from scratch and directly train a conditional generative model, *i.e.*, "*bake it in*": $p_{\theta}(x|y)$.
2. Start with unconditional model, and "*update it*" using calls to the predictive model/data (*e.g.*, CbAS [1-3], DPO [4]).
3. *On-the-fly*: freeze the unconditional model, and "*guide it*" at generation time (*e.g.*, diffusion/flow models) [5-7].

1. Brookes, Park, Listgarten *ICLM* 2019
2. Fannjiang & Listgarten *NeurIPS* 2020
3. Brookes, Busia, *et al.* *GECCO* 2020

4. Rafailov *et al.* *NeurIPS* 2023
5. Sohl-Dickstein *et al.* *ICML* 2015
6. Dhariwal & Nichol *NeurIPS* 2021
7. Song *et al.* *ICLR* 2021

You are (or should be) using Bayes rule!

For any “plug-and-play” strategy, Bayes rule is only correct operation:

$$p(x | y) = \frac{p(y | x)p(x)}{p(y)}$$

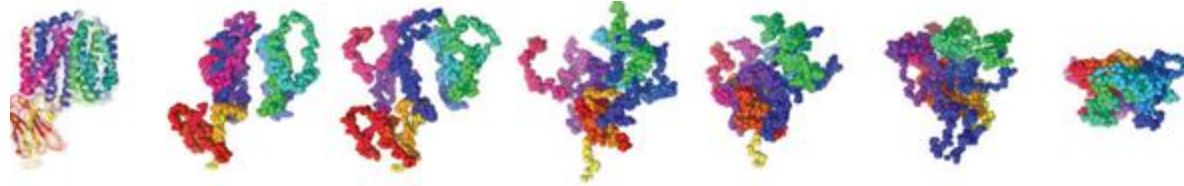
*property
predictor
e.g. CLEAN
for EC#.*

*uncondition
al model,
e.g. ESM3.*



Down Bad

The beauty of diffusion/FM/score models



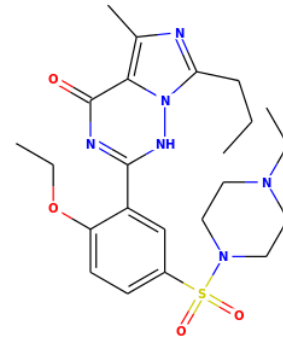
- Key diffusion model quantity: estimate $\nabla_{\mathbf{x}} p_{\theta}(\mathbf{x})$, not $p_{\theta}(\mathbf{x})$.
- By pushing gradient through Bayes rule, we get rid of nasty normalizing constant.

$$\begin{aligned}\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{y}) &= \nabla_{\mathbf{x}} \log \frac{p_t(\mathbf{x}) p_t(\mathbf{y}|\mathbf{x})}{p_t(\mathbf{y})} \\ &= \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) + \nabla_{\mathbf{x}} \log p_t(\mathbf{y}|\mathbf{x}) - \cancel{\nabla_{\mathbf{x}} \log p_t(\mathbf{y})} \\ &= \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) + \nabla_{\mathbf{x}} \log p_t(\mathbf{y}|\mathbf{x}).\end{aligned}$$

Unconditional model Guidance for conditioning

What about diffusion/flow on discrete state spaces?

MYTWTGALITPCAAEESKLPINPLSNSLLRHH
YDTRCFDSTVTESDIRVEESIYQCCDLAPEEA
LTERLYIGGPLTNSKGQNCGYRRCRASGVLT
SCGNTLTCYLKATAACRAAKLQDCTMLVNGDD
LVVICESAGTQEDAAALRAFTEAMTRYSAAPP



Challenge: for sequences, graphs, text, don't have $\nabla_x p_\theta(x)$.

Consequence I: Standard diffusion/score doesn't work.

Consequence II: Cannot use standard guidance, $\nabla_x p_\phi(y|x)$.

Some mitigating strategies:

- Relax x into real-valued space and snap back.
- Diffusion(flow) on multinomial space [Stark *et al.* *arXiv* 2024].
- Continuous-time Markov processes [Campbell *et al.* 2022, 2024]

Unlocking Guidance for Discrete State-Space Diffusion and Flow Models

Hunter Nisonoff



Junhao (Bear) Xiong

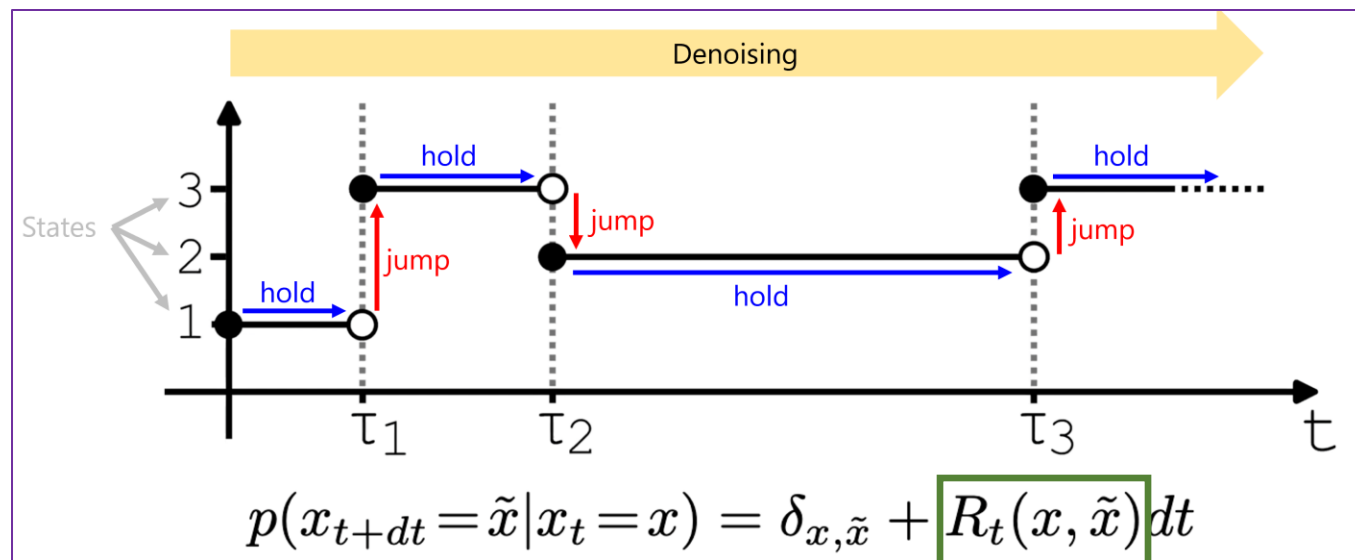


CTMC enable not only diffusion, but also guidance, on both diffusion and flow models.

Nisonoff*, Xiong*, Allenspach*, Listgarten, ICLR 2024



Stephan Allenspach



Campbell et al., NeurIPS (2022)

Campbell*, Yim* et al., ICML (2024)

Bayes Rule is easy again!

**Continuous
state-space**

$$\nabla_{x_t} \log p^{(\gamma)}(x_t|y) = \nabla_{x_t} \log \overset{\text{unconditional}}{p(x_t)} + \gamma \nabla_{x_t} \log \overset{\text{predictor}}{p(y|x_t)}$$

J. Sohl-Dickstein *et al.*, ICML (2015).

P. Dhariwal* and A. Nichol*, NeurIPS (2021).

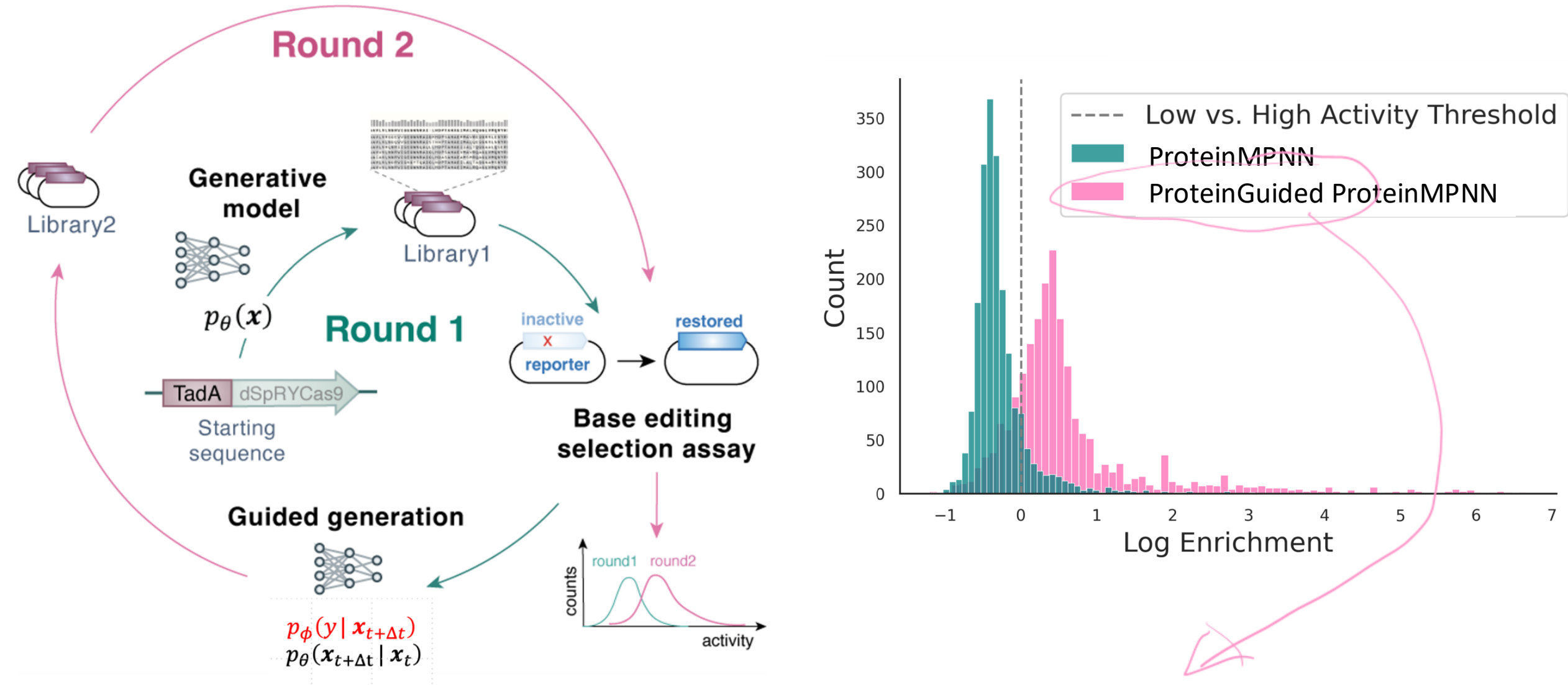
Y. Song *et al.*, ICLR (2021).

**Discrete
state-space**

$$\log R_t^{(\gamma)}(x, \tilde{x}|y) = \log \overset{\text{unconditional}}{R_t(x, \tilde{x})} + \gamma \left(\log \overset{\text{predictor}}{p(y|\tilde{x}, t)} - \log \overset{\text{predictor}}{p(y|x, t)} \right)$$

[Nisonoff*, Xiong*, Allenspach*, Listgarten, *ICLR* 2025]

ProteinGuide: experimental TadA base editor



Xiong*, Nisonoff*, Lukarska*, Gaur*, Oltrogge, Savage⁺, Listgarten⁺, *Nat. Biotech (in press)*

Engineering AAV for gene therapy delivery

The Adeno-associated virus (AAV) is a non-pathogenic virus that shows promise for delivering gene therapies (e.g. deliver blindness therapy to outer retina).

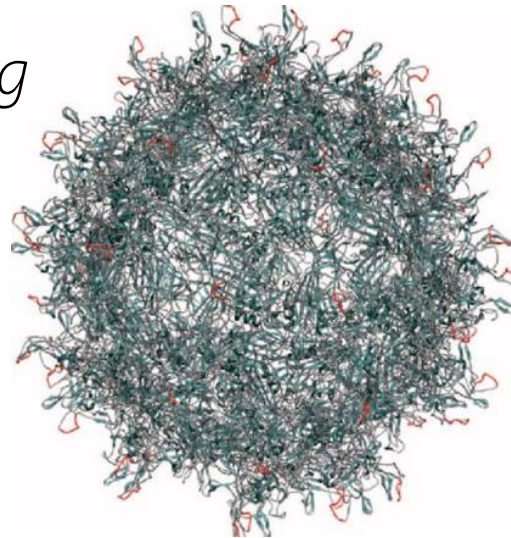
UC Berkeley: Chem. & Bio. Engineering



David Schaffer



Bonnie Zhu



David Brookes
(now at Dyno)



Akosua Busia
(now at Dyno)

[Zhu](#), [Brookes](#), [Busia](#),..., Nowakowski, [Listgarten](#), [Schaffer](#), *Science Advances* 2024

Promising AAV clinical trials

Recent clinical trial success:

Leber's congenital amaurosis (AAV)

Spinal muscular atrophy (AAV)

Hemophilia B (AAV)

Lipoprotein lipase deficiency (AAV)

Many diseases targets are still beyond the reach of current gene delivery technology

The NEW ENGLAND JOURNAL of MEDICINE

BRIEF REPORT

Safety and Efficacy of Gene Transfer for Leber's Congenital Amaurosis

FDA approves gene therapy for blindness

Health • Diet • Fitness • Living Well • Parenting • Family

Single-Dose Gene Therapy for Spinal Muscular Atrophy

Adenovirus-Associated Virus Vector-Mediated Gene Transfer in Hemophilia B

J.R. Mendell, S. Al-Zahrani, J.W.D. Arnold, L.R. Rodino-Klapac, T.W. Prior, L. Lowes, L. Alfano, K. Berry, K. Church, J.T. Kissel, S. Nagendran, J. L'Italien, D.M. Sproule, C. Wells, J.A. Cardenas, M.D. Heitzer, A. Kaspar, S. Corcoran, L. Braun, S. Likhite, C. Miranda, K. Meyer, K.D. Foust, A.H.M. Burghes, and B.K. Kaspar

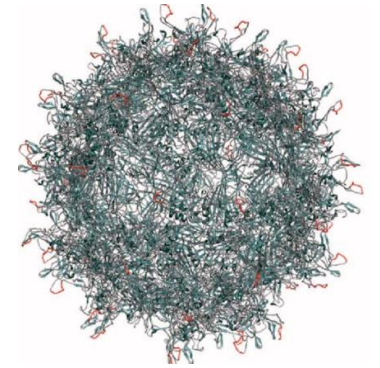
The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 DECEMBER 22, 2011 VOL. 365 NO. 25

Amit C. Nathwani, M.B., Ch.B., Ph.D., Edward G.D. Tuddenham, M.B., B.S., M.D., Savita Rangarajan, M.B., B.S., Cecilia Rosales, Ph.D., Jenny McIntosh, Ph.D., David C. Linch, M.B., B.Chir., Pratima Chowdary, M.B., B.S., Anne Riddell, B.Sc., Arnulfo Jaquilmac Pie, B.S.N., Chris Harrington, B.S.N., James O'Beirne, M.B., B.S., M.D., Keith Smith, M.Sc., John Pasi, M.D., Bertil Glader, M.D., Ph.D., Pradip Rustagi, M.D., Catherine Y.C. Ng, M.S., Mark A. Kay, M.D., Ph.D., Junfang Zhou, M.D., Yunyu Spence, Ph.D., Christopher L. Morton, B.S., James Allay, Ph.D., John Coleman, M.S., Susan Sleep, Ph.D., John M. Cunningham, M.D., Deokumar Srivastava, Ph.D., Etiena Basner-Tschakarjan, M.D., Federico Mingozzi, Ph.D., Katherine A. High, M.D., John T. Gray, Ph.D., Ulrike M. Reiss, M.D., Arthur W. Nienhuis, M.D., and Andrew M. Davidoff, M.D.

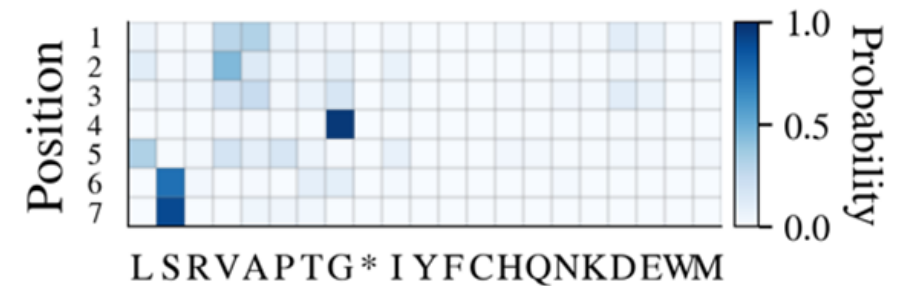
Ongoing challenges for AAV-based therapeutics

- Inefficient delivery to target tissues/cells.
- Non-specific delivery.
- Pre-existing immunological neutralization.
- Inefficient uptake into target cells.

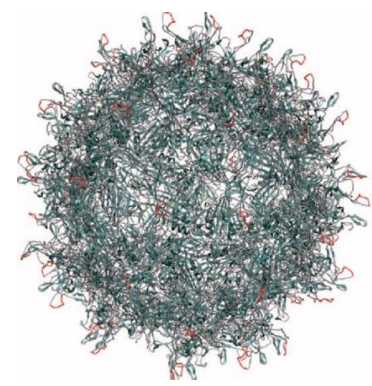


First AAV project goal, "library design":

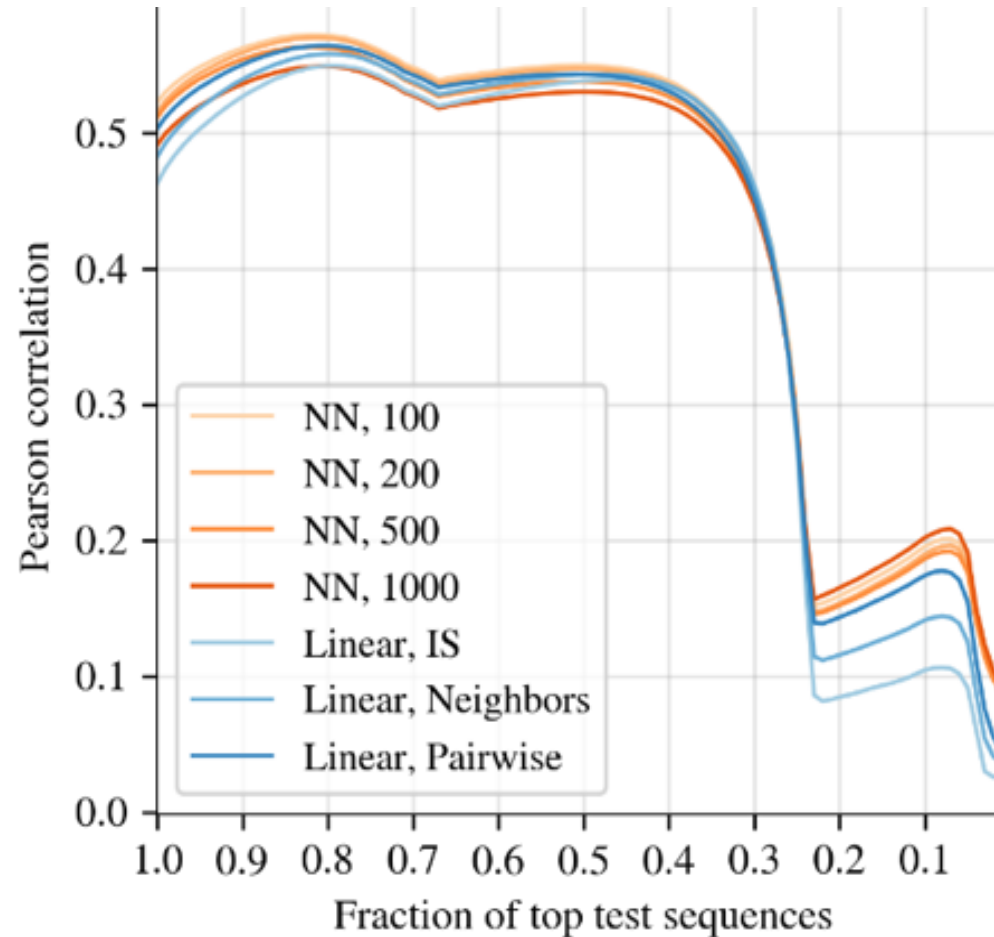
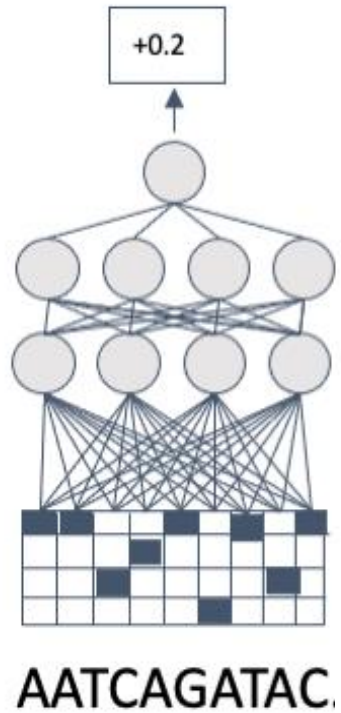
- Obtain optimal starting “library” for all these engineering goals.
- *i.e.*, large fraction of library gets wasted because doesn't *package*.



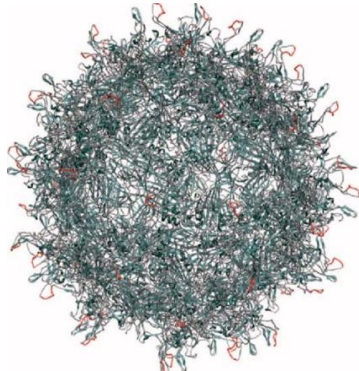
AAV library design



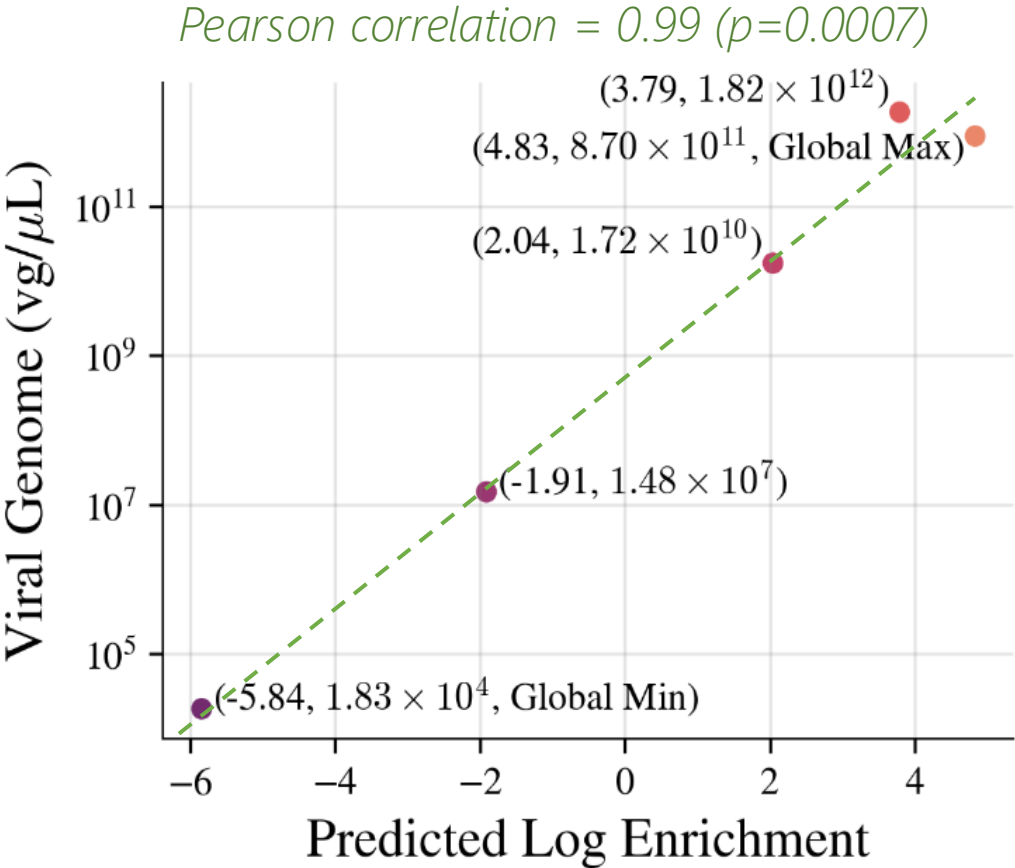
1. Build predictive model and test (*sequence*→*packaging* fitness).



AAV library design

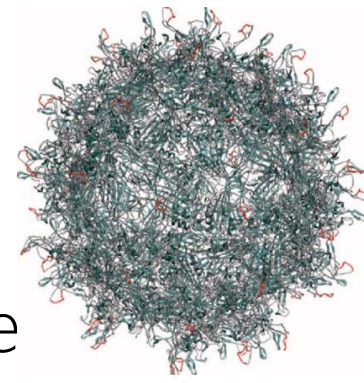


2. Wetlab **validate model** (measure titer directly)



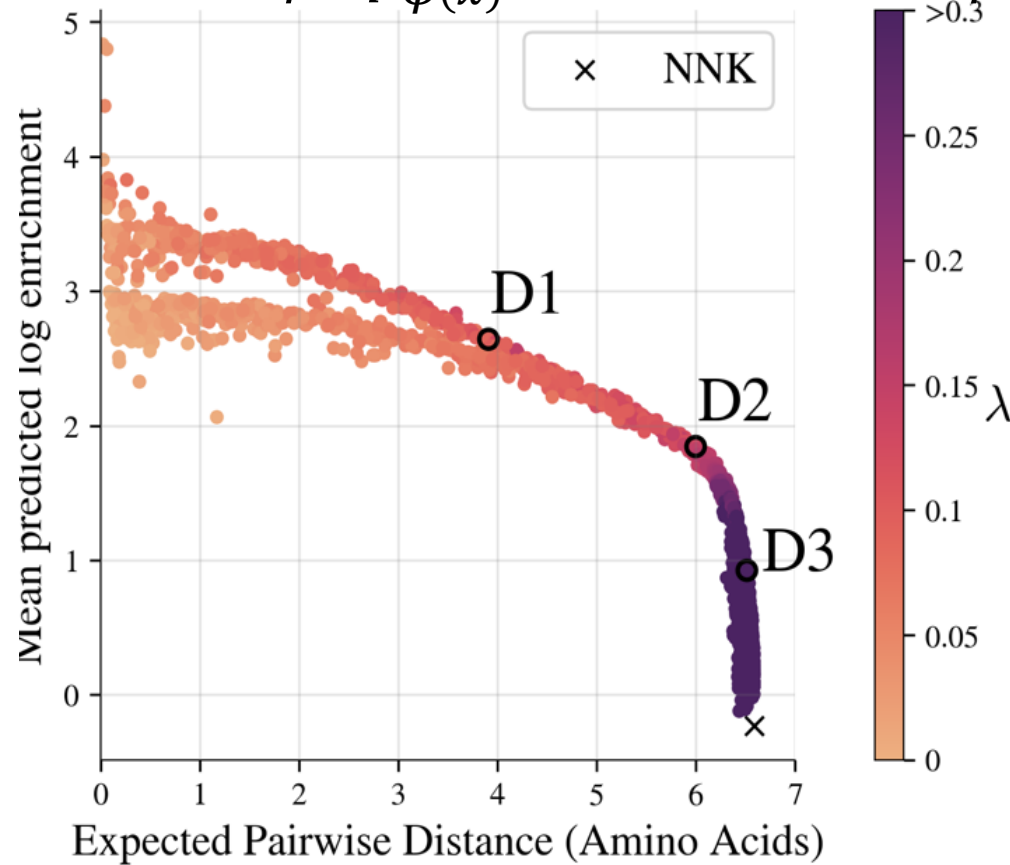
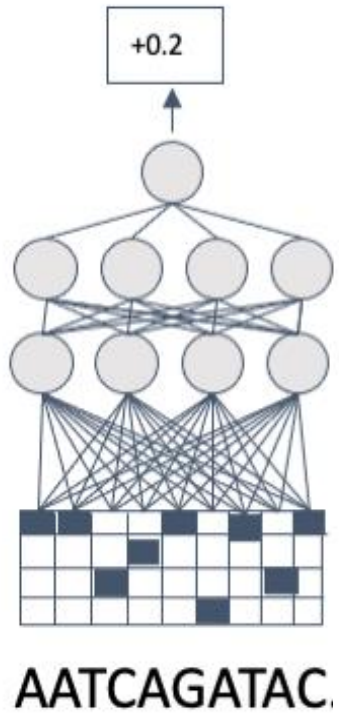
Sequences	Predicted Log Enrichment	Experimental Viral Titer (vg/ μ L)
LSSTTAA	4.834	8.70×10^{11}
DSRLSGT	3.793	1.82×10^{12}
LEPDAAL	2.044	1.72×10^{10}
IRWRATG	(-) 1.91	1.48×10^7
RWPRRVL	(-) 5.84	1.83×10^4

AAV library design



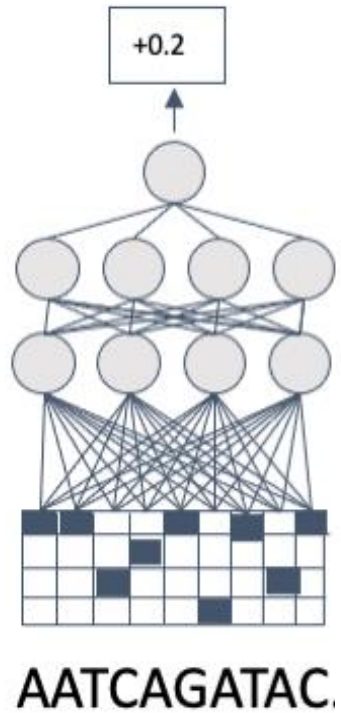
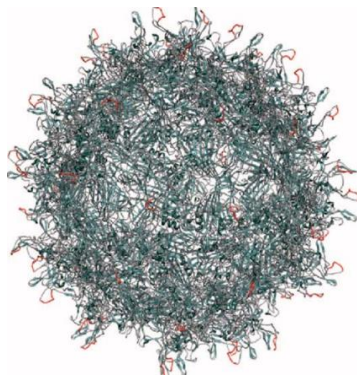
3. Invert ML predictive model to get diversity-fitness optimality curve

$$\operatorname{argmax}_{\phi} \mathbb{E}_{p_{\phi}(x)} [f(x)] + \lambda H[p_{\phi}]$$

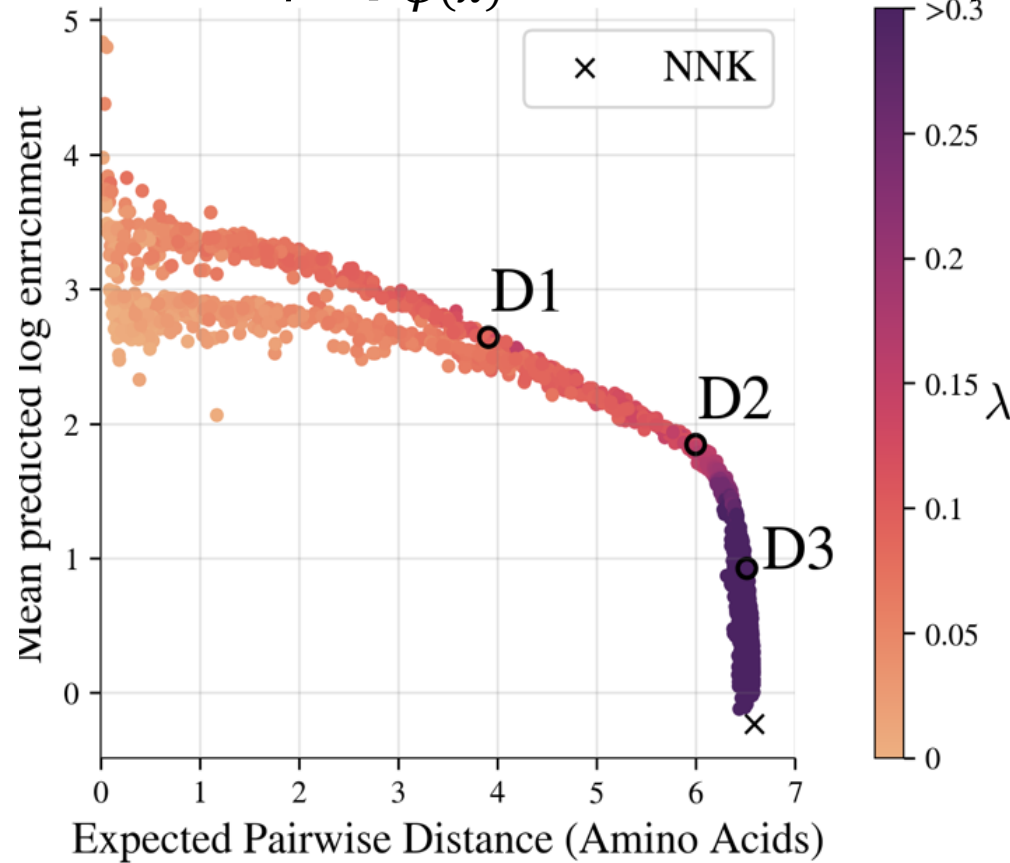


AAV library design

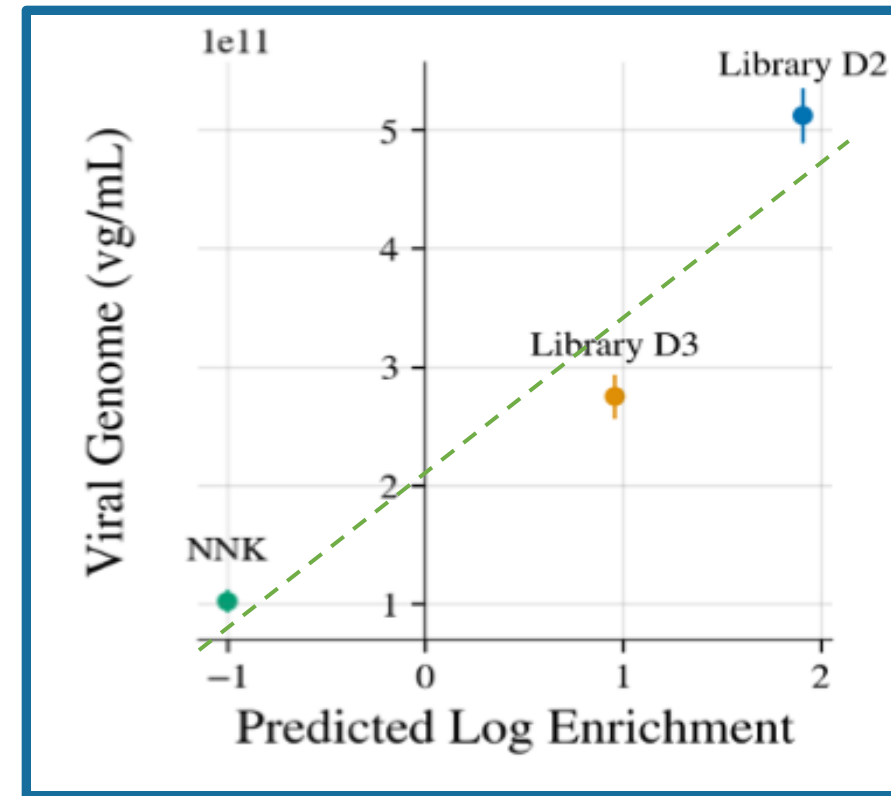
4. Validate in the lab.



$$\operatorname{argmax}_{\phi} \mathbb{E}_{p_{\phi}(x)} [f(x)] + \lambda H[p_{\phi}]$$

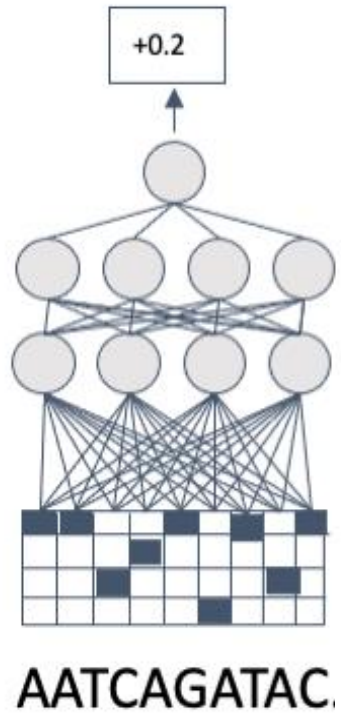
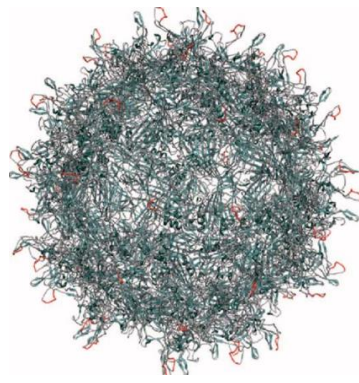


Pearson correlation = 0.96 ($p=0.001$)

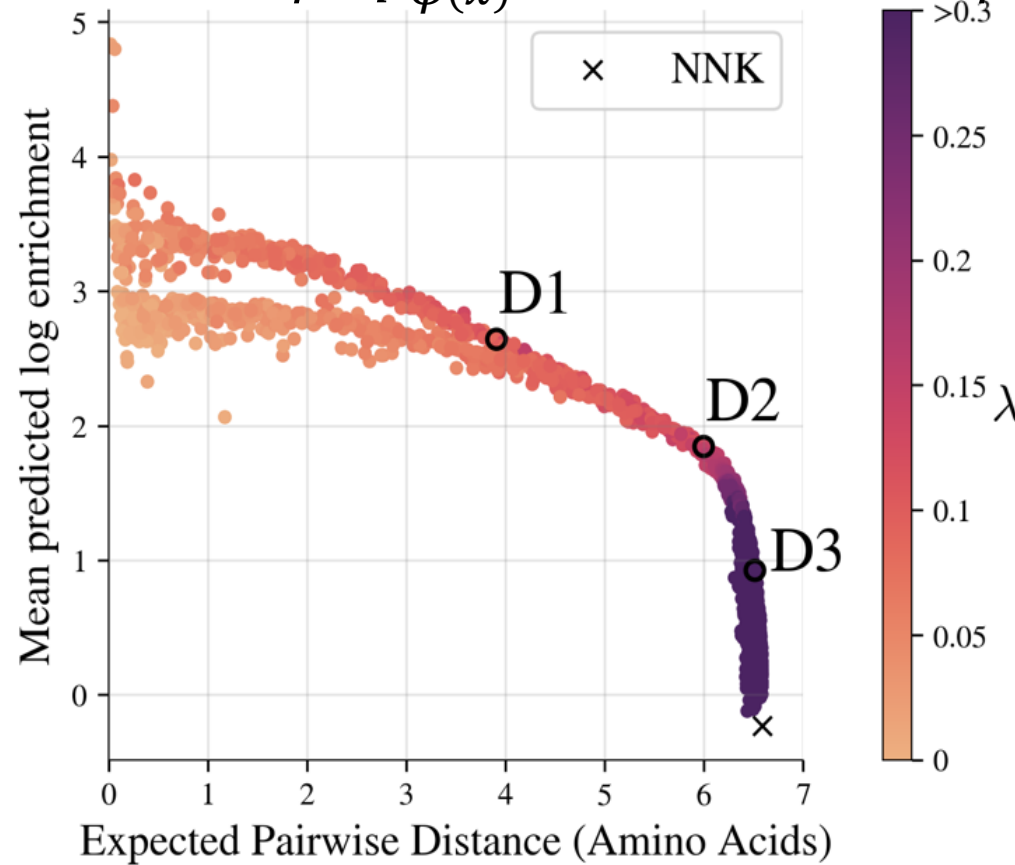


AAV library design

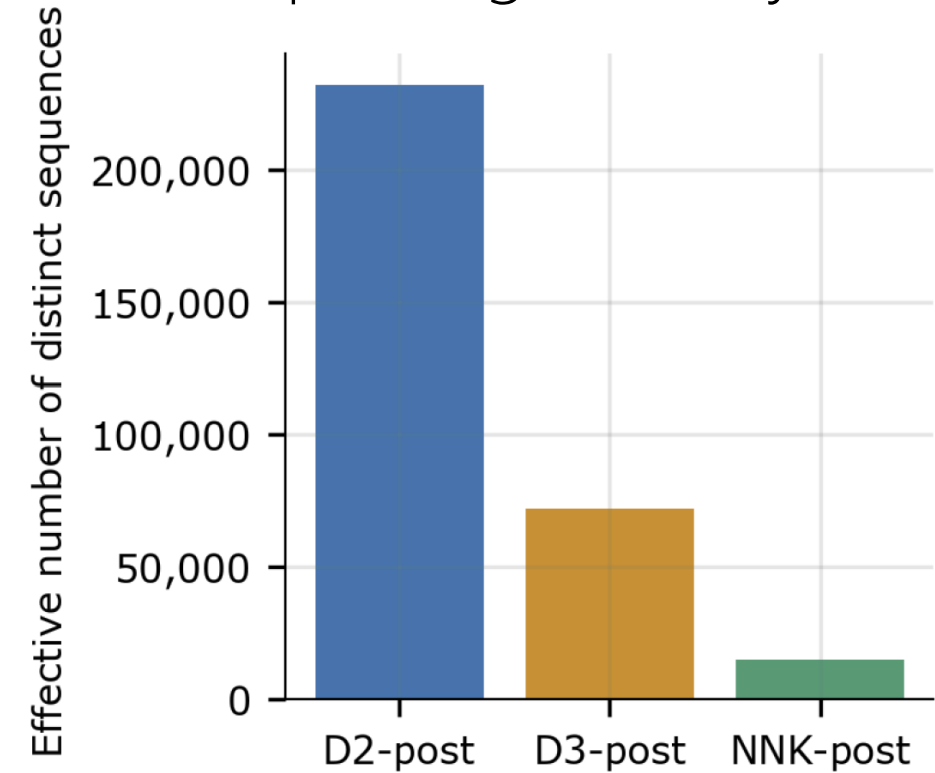
4. Validate in the lab.



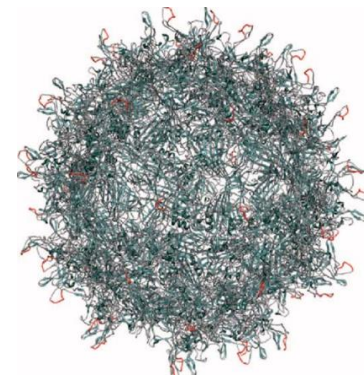
$$\operatorname{argmax}_{\phi} \mathbb{E}_{p_{\phi}(x)} [f(x)] + \lambda H[p_{\phi}]$$



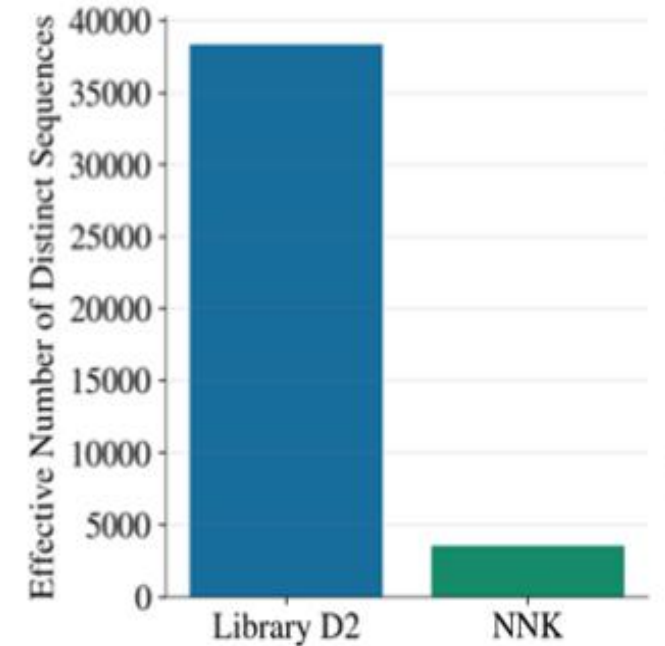
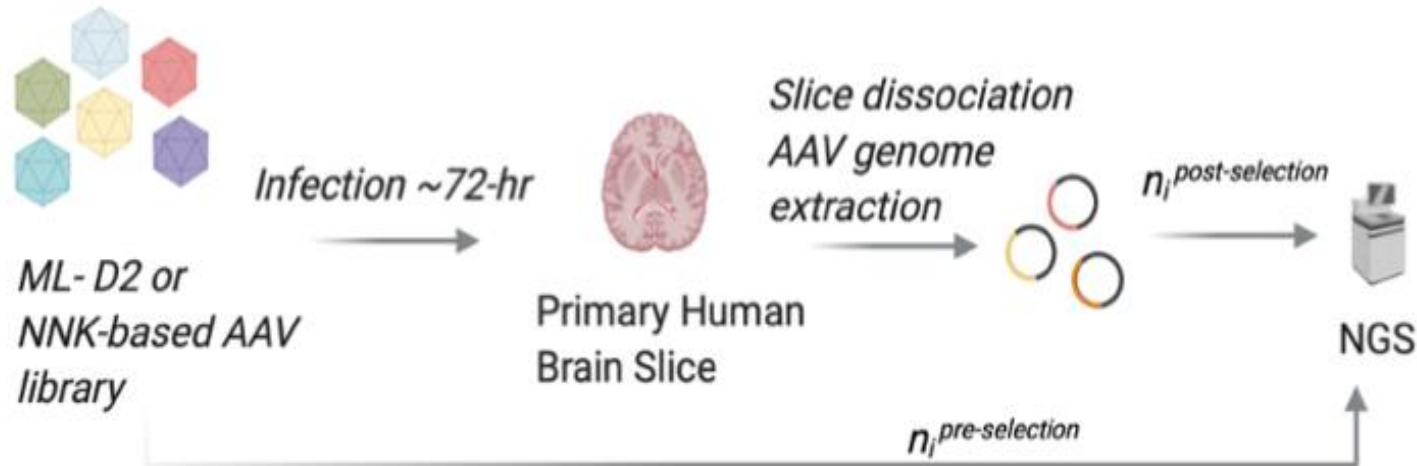
Post-packing diversity



AAV library design



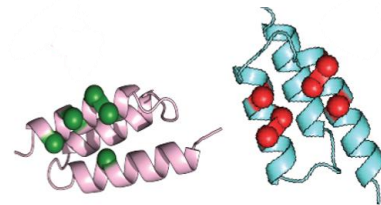
5. Demonstrate better downstream selection (human brain cell infectivity), that it *was not specifically designed for*.



*ML
library*

*currently
used
library*

The team



Christopher Garcia

Aerin Yang*

Kevin M. Jude
(*Stanford*)



Jennifer Listgarten

Hanlun Jiang*

Stephan Allenspach*

James Bowden
(*UC Berkeley*)



Tanja Kortemme

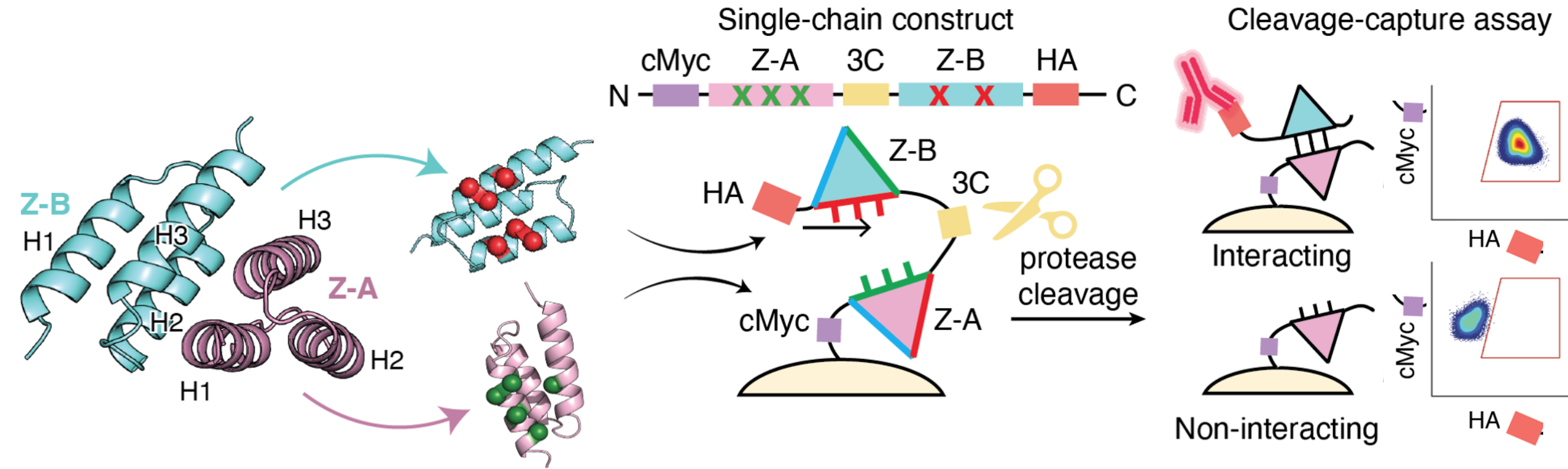
Deniz Akpınaroglu
Alex Jie Li
(*UCSF*)

Possu Huang
(*Stanford*)



(*No pre-print—in revision.*)

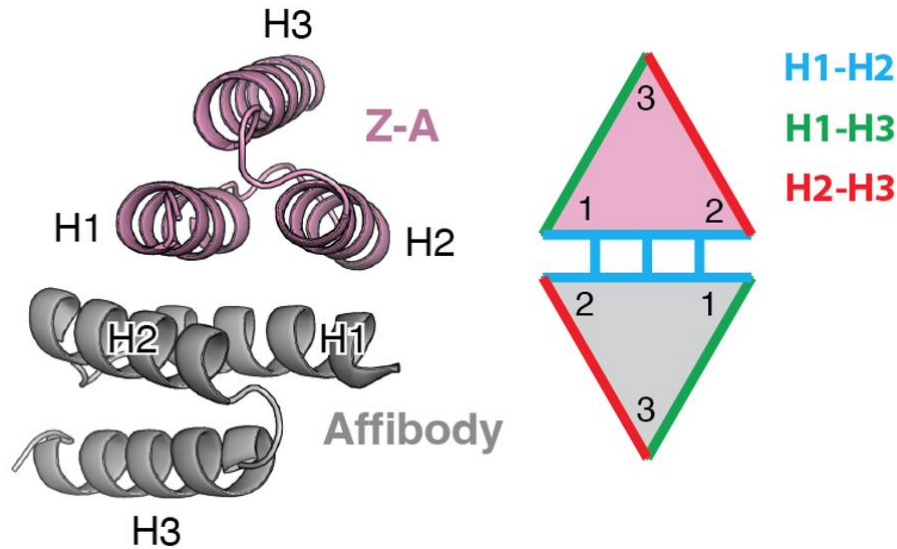
Recent platform for screening protein library on protein library (Garcia lab)



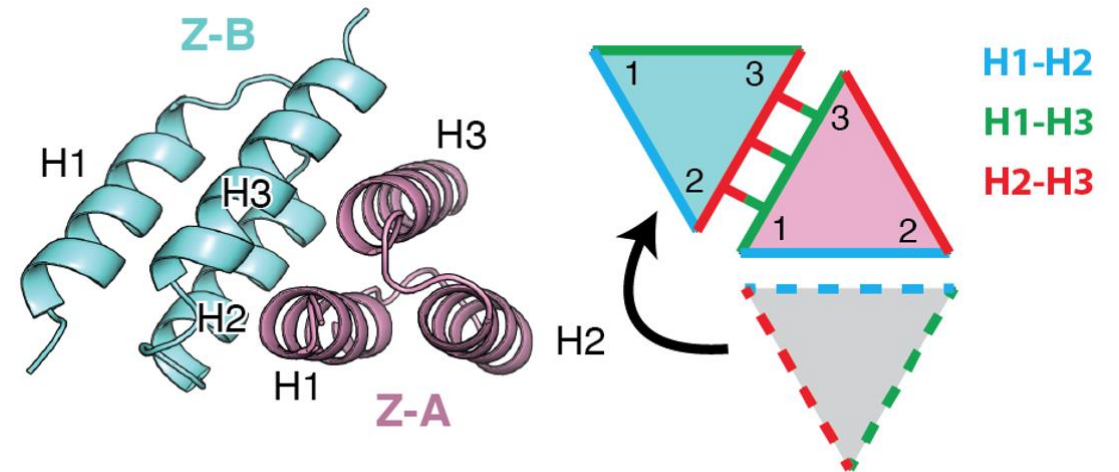
Yang, A., et al. (2023). *Science*, 381(6656)

From natural to synthetic protein binding

Natural interface



Synthetic interface



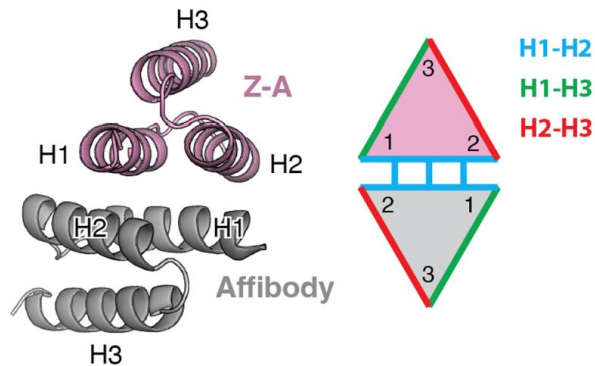
Yang, A., et al. (2023). *Science*, 381(6656)

new library

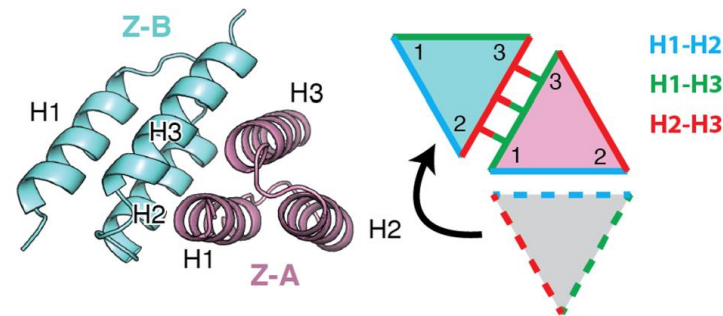
Some questions we can now ask about protein-protein interactions:

1. Are there “seeds” of binding interaction during evolution—small hints of weak binders?
2. What role does epistasis play (within vs across partners)?
3. Are synthetic and natural interfaces somehow “different”?

Natural interface

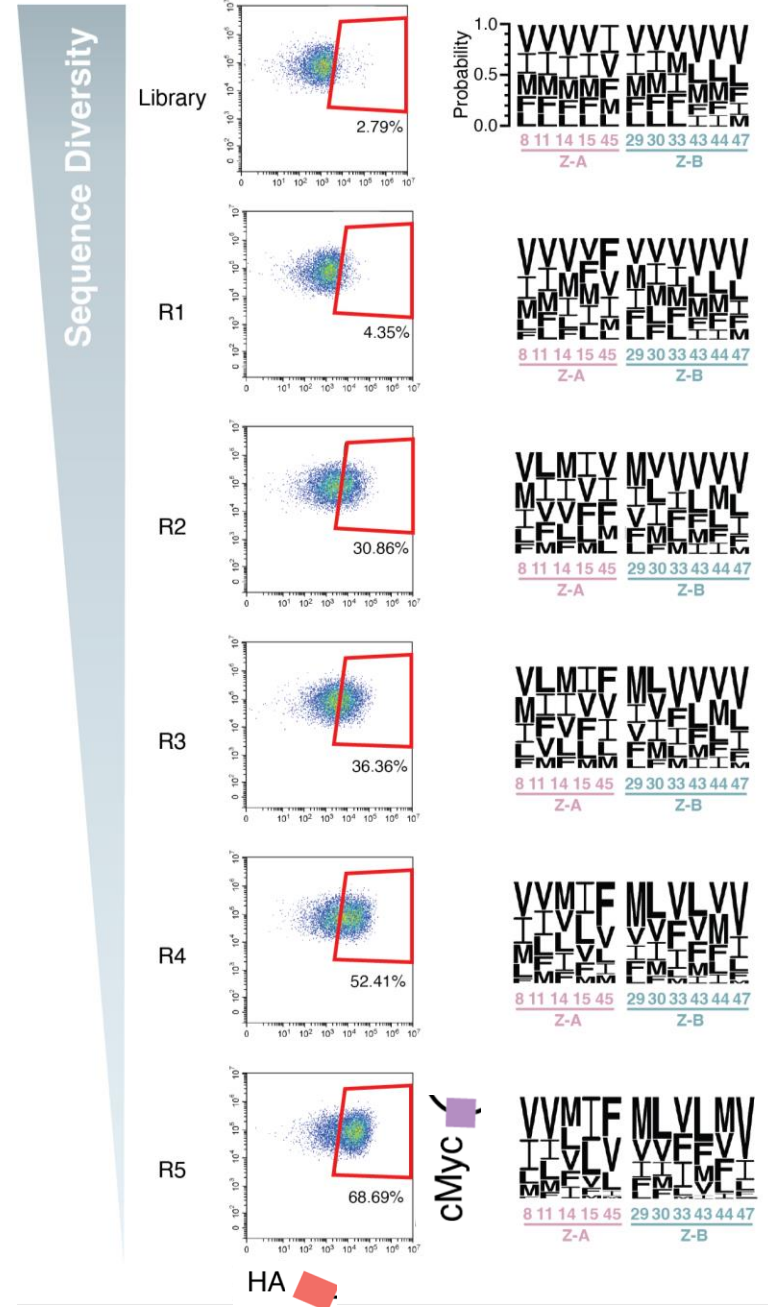
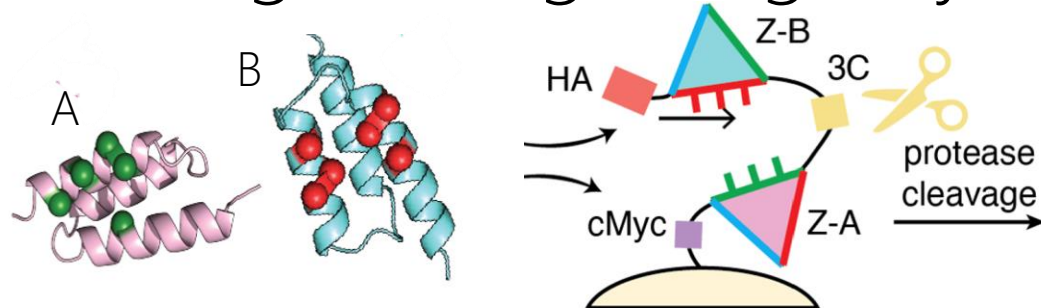


Synthetic interface

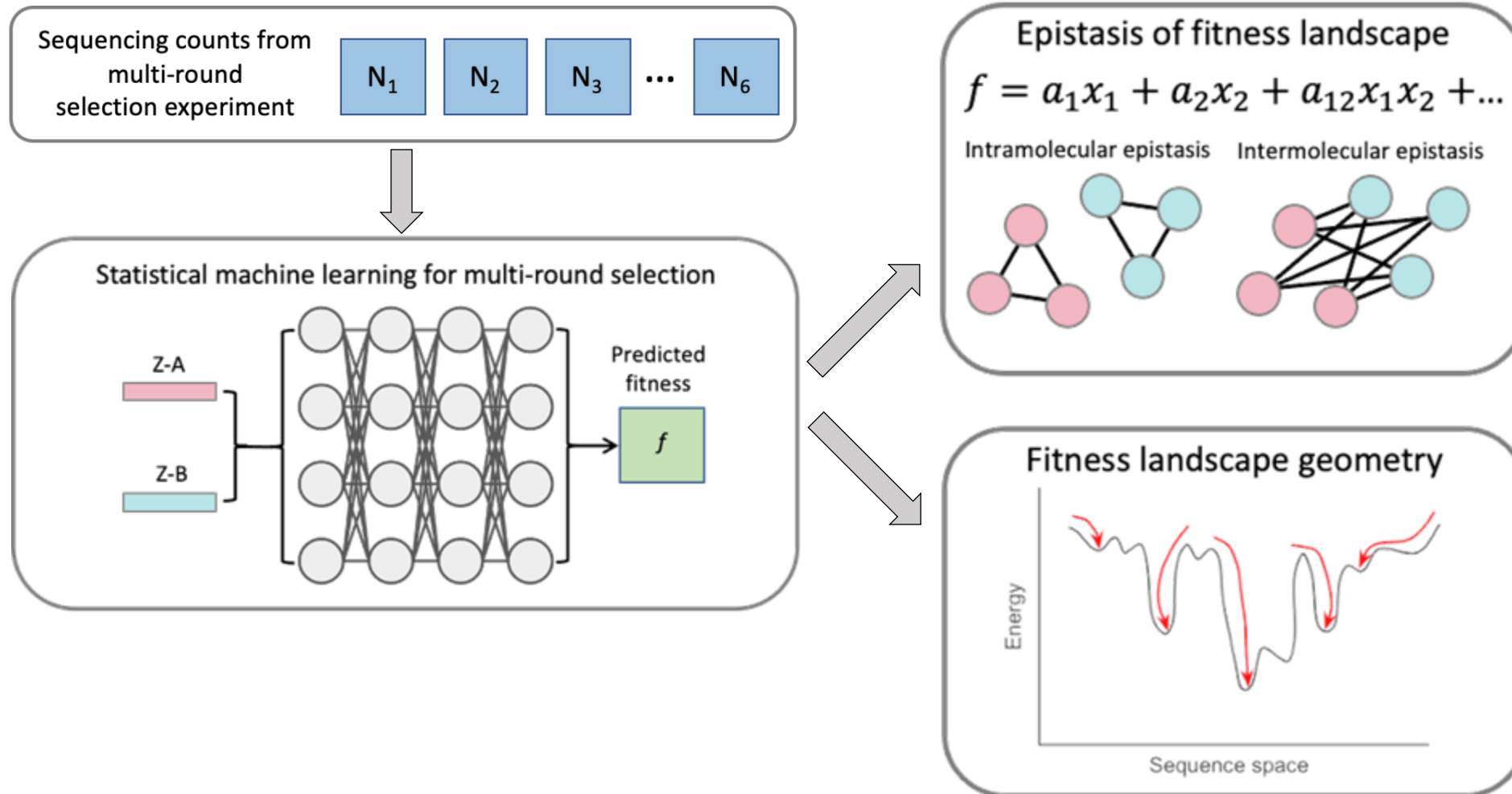


Experimental setup

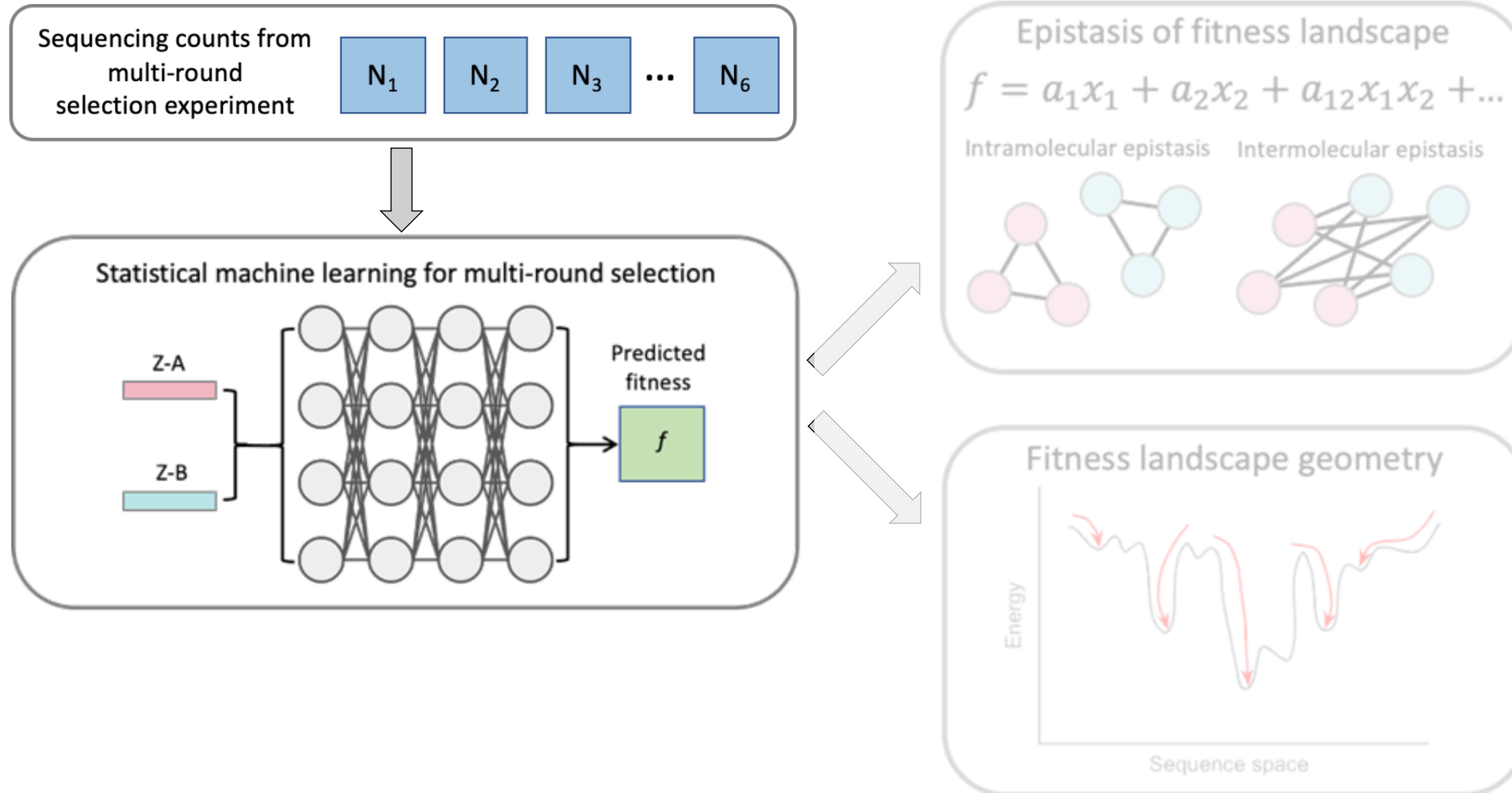
- 5 positions mutated on protein A.
- 6 positions mutated on protein B.
- Same five amino acids allowed at each position (all hydrophobic).
- Initial library has all combinations.
- $5^{6+5} = 5^{11} = 5 \times 10^7$ variants
- Five rounds of selection each with increasing binding stringency.



Overview: a statistical machine learning approach to study protein-protein coevolution



Multi-round probabilistic selection model

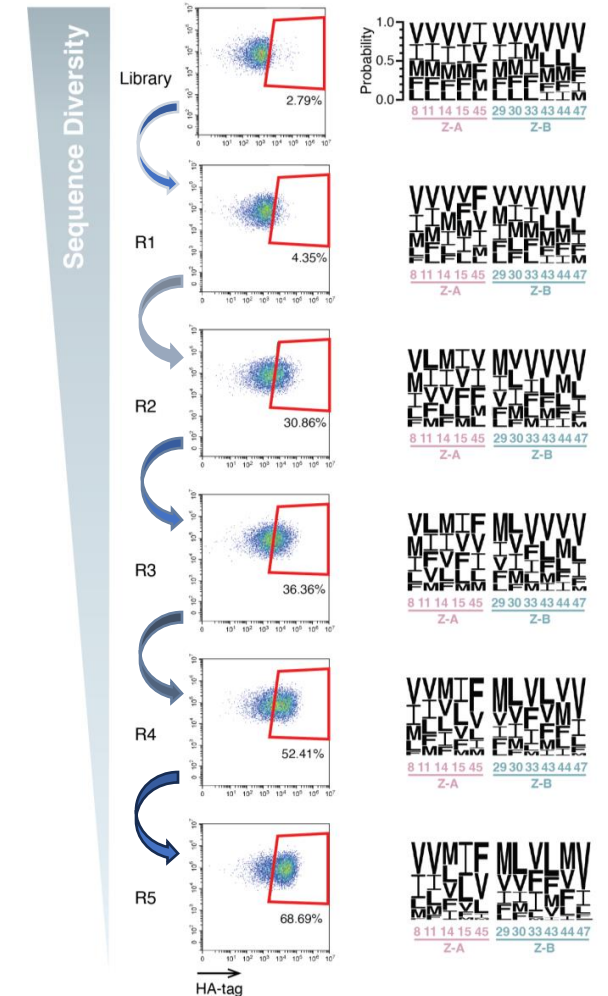
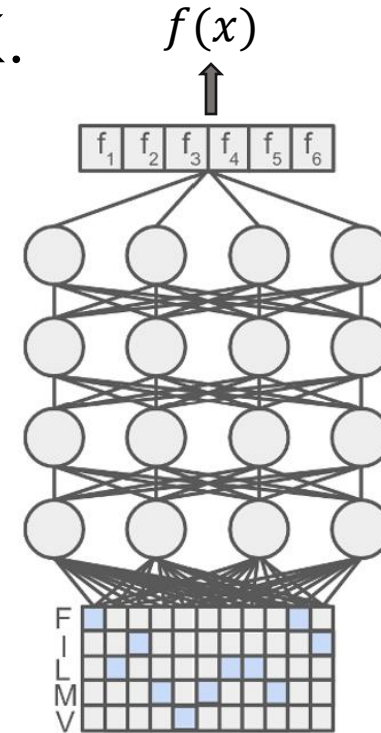
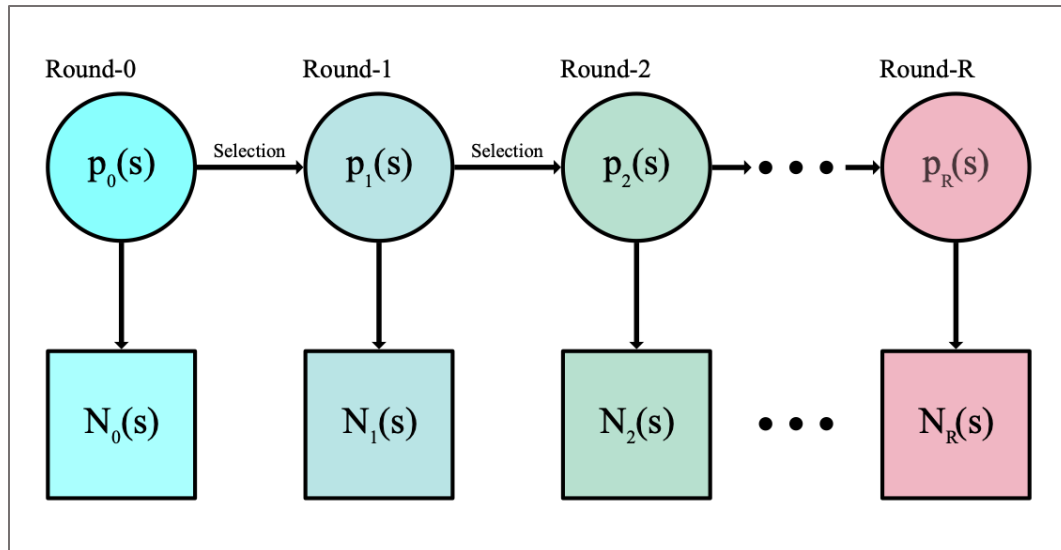


Multi-round probabilistic selection model



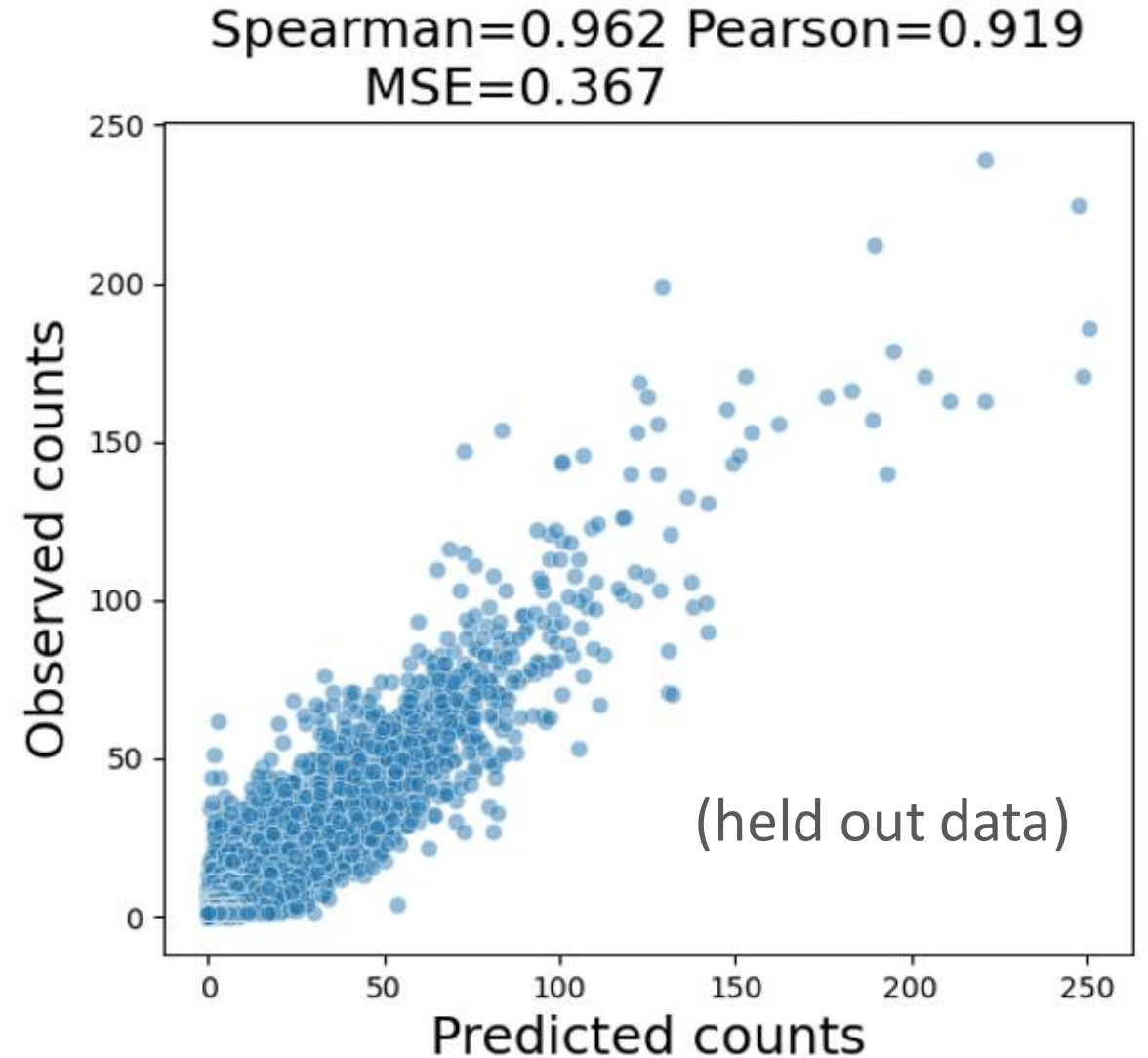
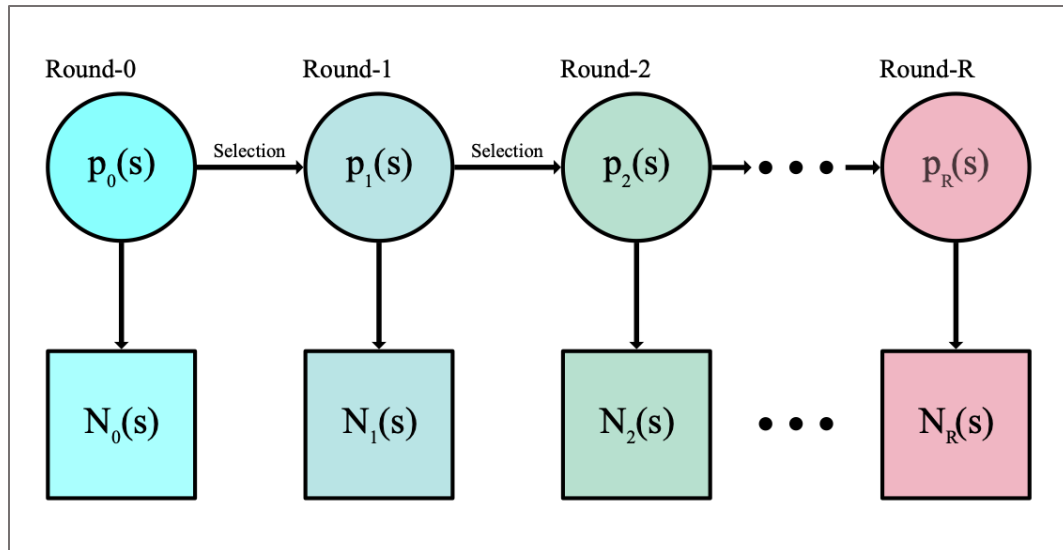
Stephan Allenspach

- Multi-round selection statistical ML model for read count data.
- Binding “*Fitness*” = $-Energy$ estimated by underlying neural network.

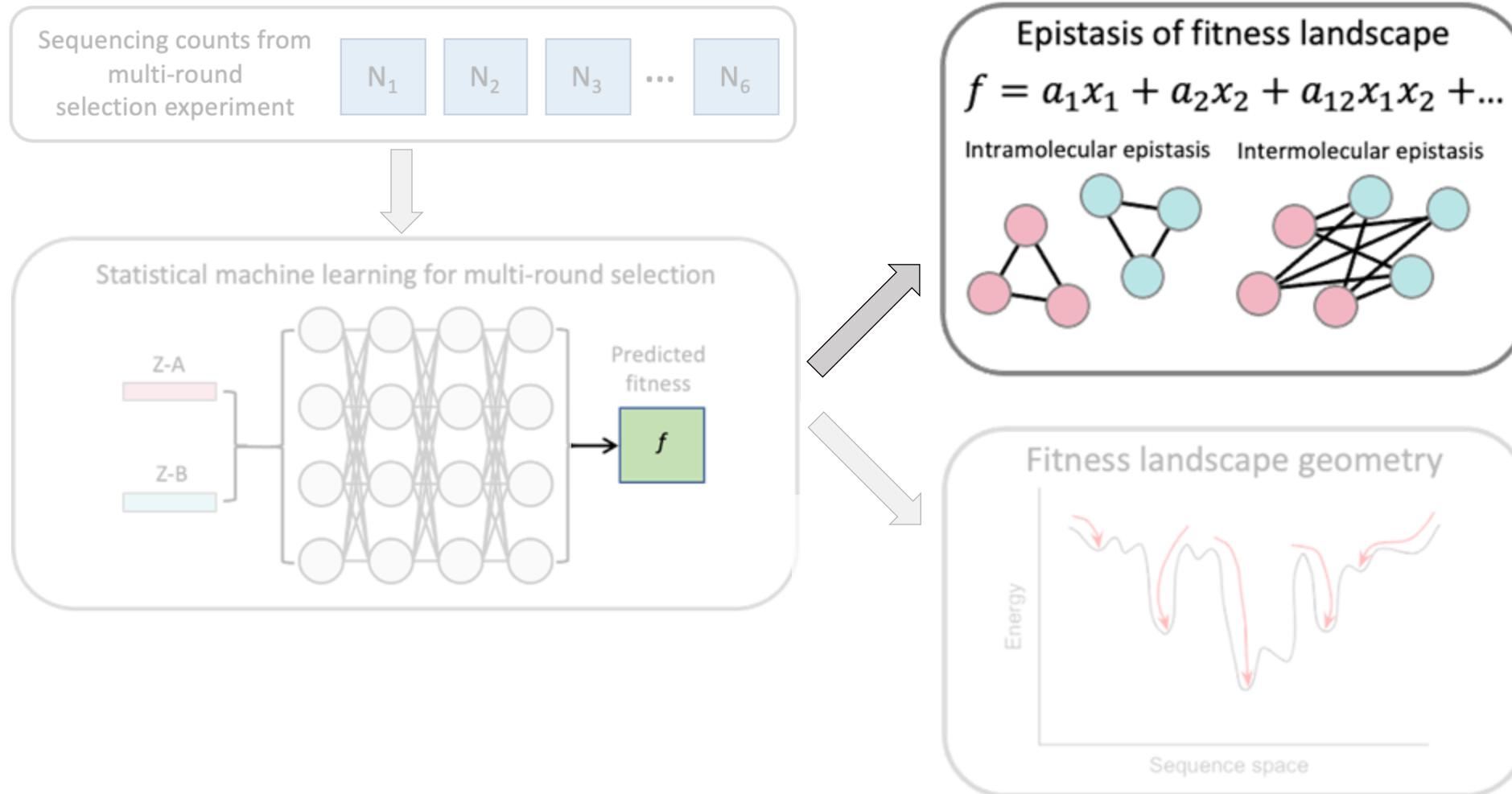


Multi-round probabilistic selection model

- Multi-round selection statistic for read count data.
- Binding “*Fitness*” = $-Ener$ by underlying neural network



Epistatic analysis of the binding fitness landscape



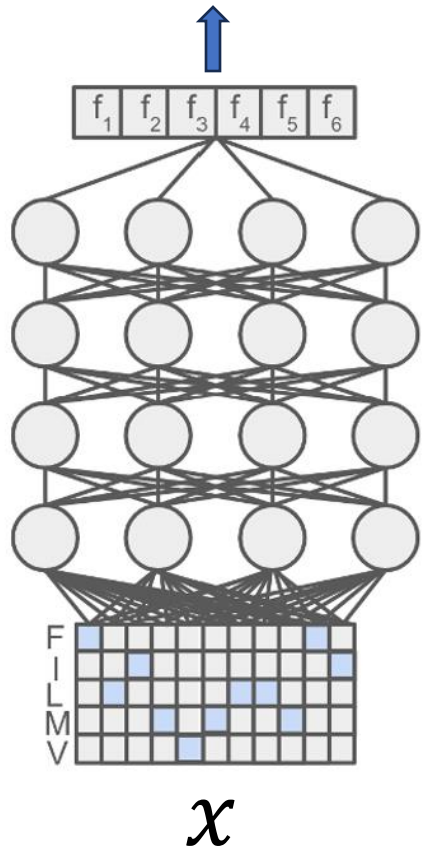
Epistatic analysis of the binding fitness landscape

–Energy
= fitness
= $f(x)$

$$f(x) = a_1x_1 + a_2x_2 + a_{12}x_1x_2 + \dots$$

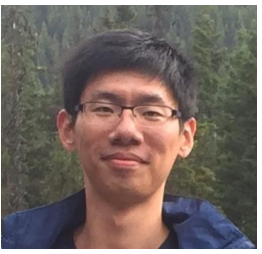


StephanAllenspach

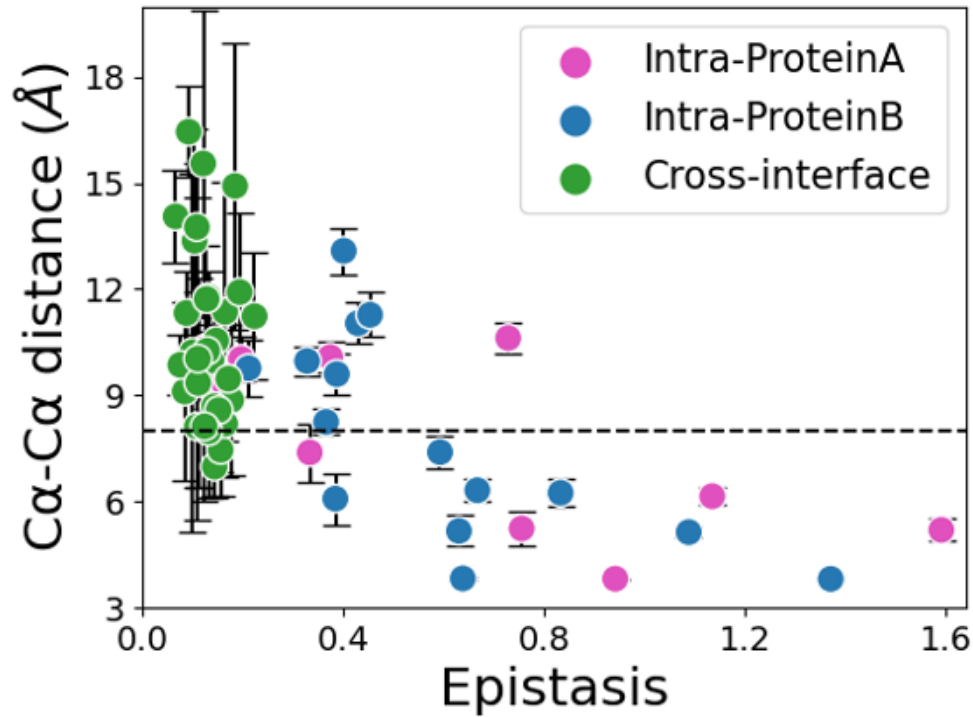


- Any fitness function can be exactly written as linear additive in all epistasis terms up to order L .
- Extracting all epistatic terms is tantamount to doing a spectral transformation of $f(x)$ e.g., [1].
- Naively time complexity $O(A^{2L})$.
- Our method $O((2A - 1)^L)$
- Also can do partial transform only up to order K more efficiently $O(A^L + \binom{L}{K} (2A - 1)^K)$.

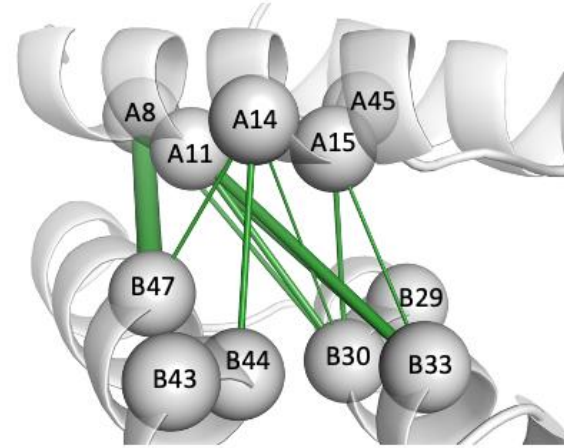
Strength of pairwise epistasis scales inversely with contact distance in crystal structure



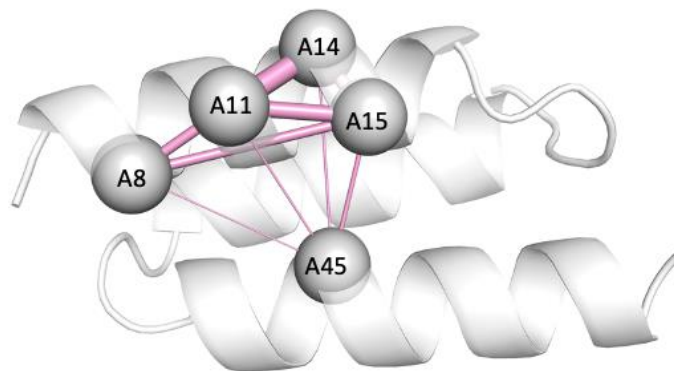
Hanlun Jiang



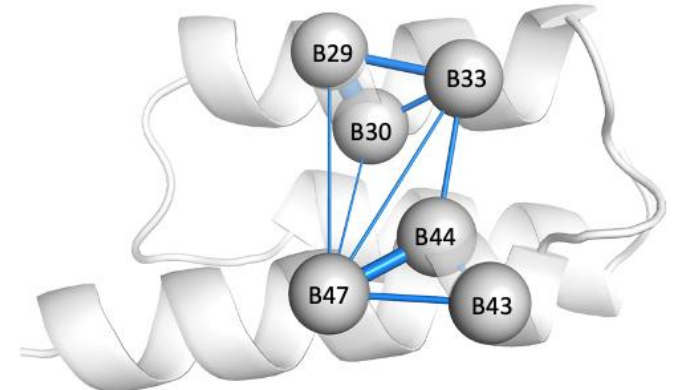
Cross-interface epistasis



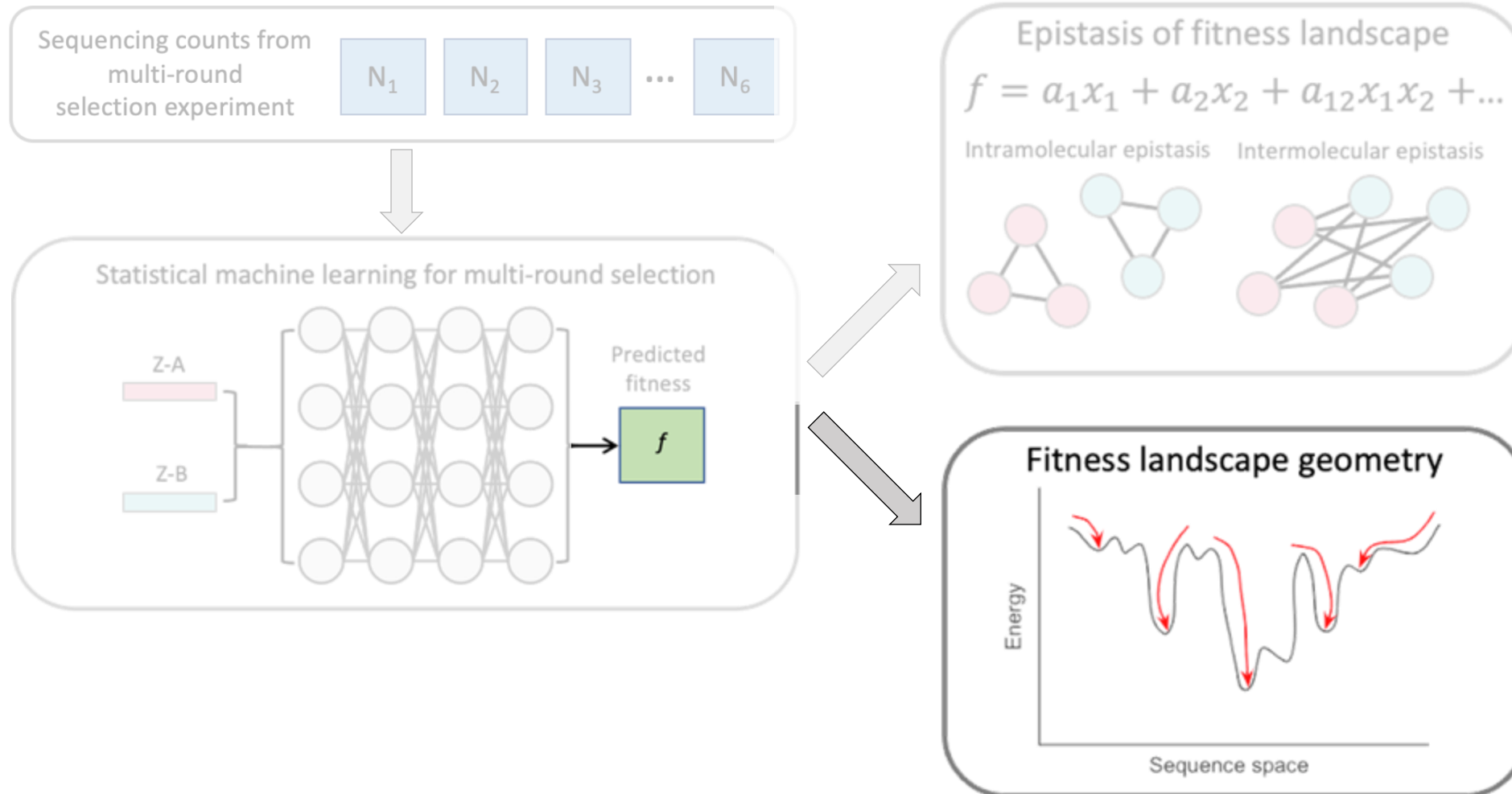
Intra-ProteinA epistasis



Intra-ProteinB epistasis

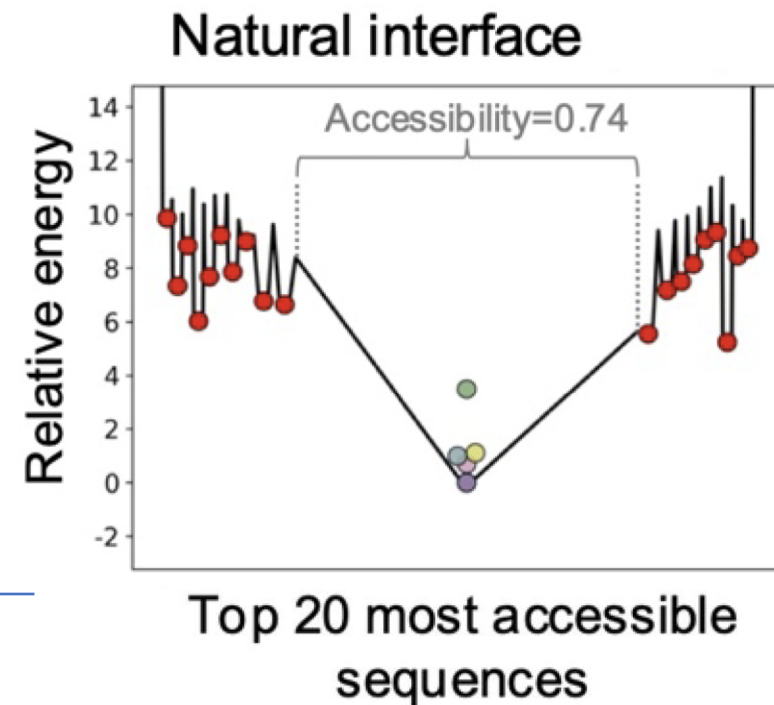
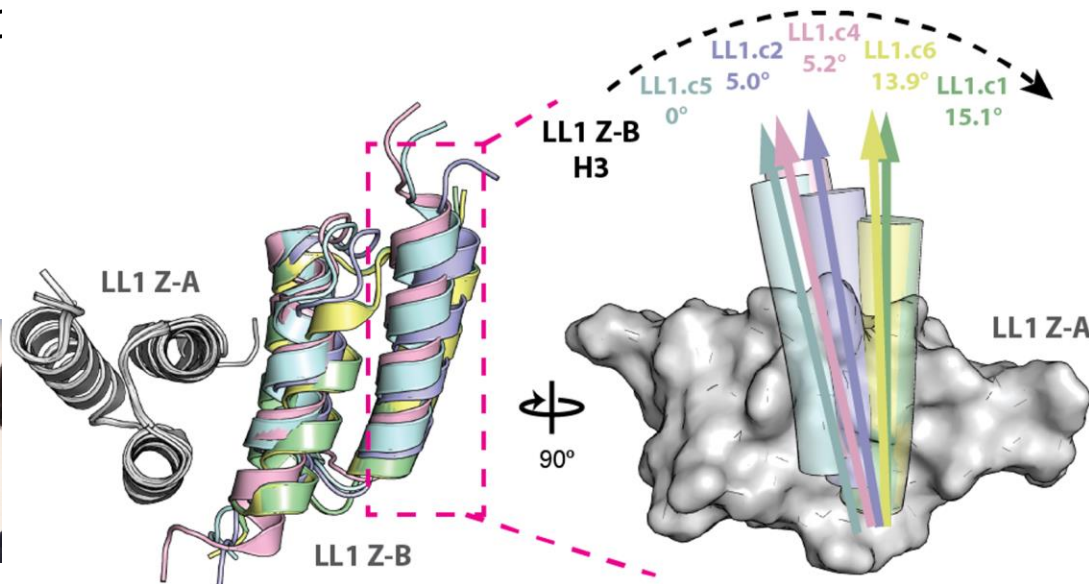
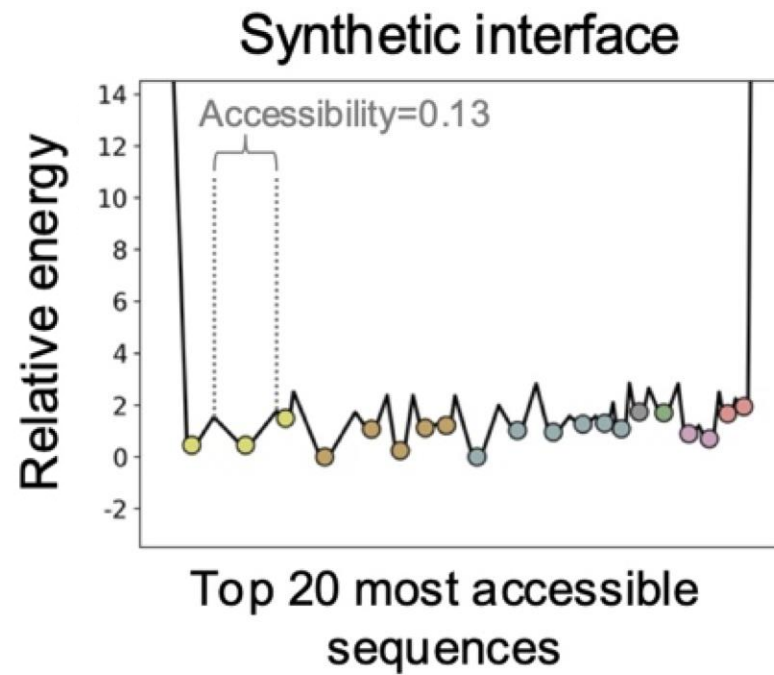
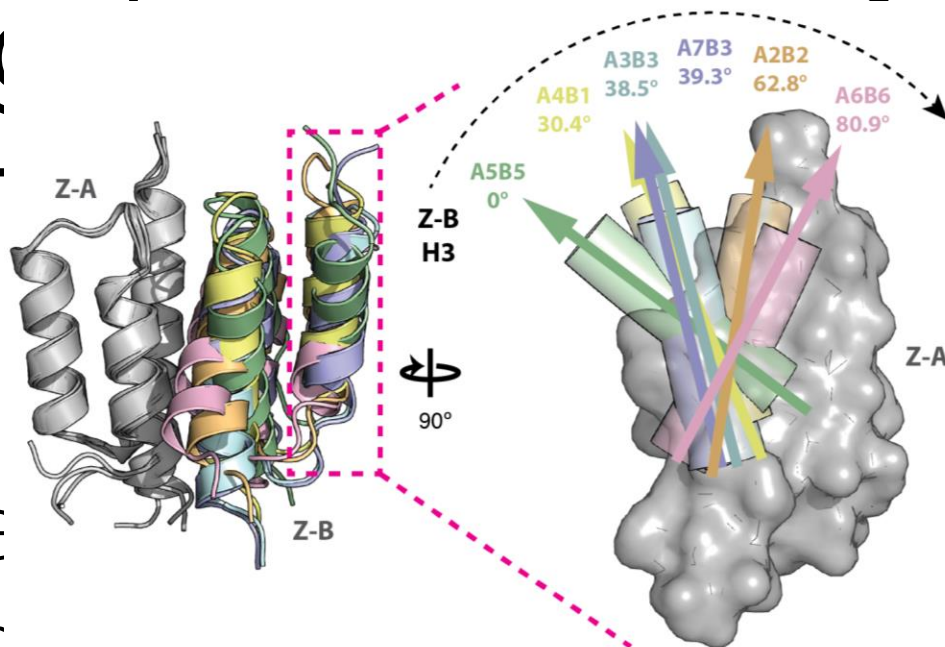


Fitness landscape geometry: natural vs synthetic



Characterizing the binding fit

- Simulated ~4M increasing fitness
- "Accessibility" of a sequence pair: fraction of α stopped at



Hanlun Jiang James Bowden Stephen J. Hupac